Interpretation of high-order harmonic generation in terms of transitions between quantum Volkov states

J. Gao, F. Shen, and J. G. Eden
Department of Electrical and Computer Engineering, University of Illinois, Urbana, Illinois 61801
(Received 19 April 1999; revised manuscript received 22 December 1999; published 20 March 2000)

A theoretical perspective of high-order harmonic generation is presented in which transitions between quantum Volkov states—stationary solutions to Schrödinger's equation for an electron interacting with a quantized optical field—are the dominant mechanism for producing radiation. The picture proposed here is one in which the interaction of an intense optical field with an electron forms a photon-electron ''quasiatom'' and the stationary states of this complex can be viewed as the high optical field analog of bound states of an atom. Consequently, although momentum conservation for the overall harmonic generation process requires that an electron eventually return to the parent atom, the electron and fundamental optical field are treated as an isolated system with well-defined energy and momentum. To introduce a secondary photon (harmonic) field, the matrix element describing the electron-photon field interaction through quantum Volkov states is modified in a perturbative manner. The result is the production of harmonic photons by Volkov→Volkov-state transitions, a process which has no classical counterpart. The generation of odd harmonics of the fundamental field and the requirement that the final state of the electron be the initial bound state of the atom follow directly from the theory. A clear connection between harmonic generation and above-threshold ionization (ATI) is drawn by maintaining the discrete, nonzero energies of photoelectrons injected into the driving optical field by ATI. For experiments in which the laser pulse width is $\gtrsim$100–150 fs, only the lowest ATI photoelectron orders contribute significantly to harmonic generation, whereas higher-energy electrons escape from the parent atom during the pulse risetime. As the duration of the fundamental field is decreased, however, the rapid rise of the field intensity to its peak value effectively traps more energetic photoelectrons, thereby extending the harmonic response to shorter wavelengths. An analytic expression for the transition matrix for the harmonic generation process is derived and is compared with the predictions of semiclassical theory and published experimental results, including harmonic spectra and polarization data.

PACS number(s): 42.65.Ky, 42.50.Hz, 32.80.Rm

I. INTRODUCTION

It has been known for more than three decades that odd harmonics of the optical field frequency are generated when an intense laser pulse interacts with an atom in the gas phase. In 1967, New and Ward [1] reported producing the third harmonic of the ruby laser in the rare gases and the ensuing years have witnessed steady improvements in the number of harmonics produced. This progress in generating ever-decreasing wavelengths mirrors advances in laser technology that continue to reduce pulse widths while maintaining pulse energies in the mJ range. Early experiments by Reintjes and colleagues [2] in He and Ne produced harmonics having wavelengths as short as 38 nm with ~30-ps pulses at 266 nm (fourth harmonic of Nd:YAG) and, following the introduction of the excimer laser, several studies [3,4] were carried out at 308 nm (XeCl) and 248 nm (KrF) with pulse widths as low as 15 ps (Ref. [4]). In one experiment, coherent radiation at the seventh harmonic of 248 nm (35.5 nm) was observed.

A turning point in the field occurred when McPherson *et al.* [5] detected radiation corresponding to the 17th harmonic of 248 nm when Ne was exposed to driver field intensities in the 10^{15}–$10^{16}\,\mathrm{W\,cm^{-2}}$ range. More importantly, the rate of decline of the relative intensity of the harmonics slowed dramatically beyond the 11th harmonic. Similar results were observed for the other rare gases and Ferray and co-workers [6] and Li *et al.* [7] of Saclay subsequently confirmed the presence of higher-order harmonics (harmonic order $q \leqslant 33$ for Ne) for optical fields of longer wavelength as well ($\lambda = 1.06\,\mu\mathrm{m}$). An intriguing characteristic of the harmonic spectra reported in Refs. [5–7] is a region, known as the plateau, in which the harmonic intensity is relatively insensitive to q, followed by cutoff, a rapid decline in the harmonic response. For the *lowest* harmonics ($q \lesssim 7$), intensity also falls rapidly with increasing harmonic order (as had been observed previously), prompting Li *et al.* [7] to comment that ''The first harmonics behave . . . according to theoretical predictions derived from lowest-order perturbation theory . . . .'' However, '' . . . neither the extended plateau . . . nor the cutoff at high order can be explained by the traditional theory of nonlinear optics in [*a*] weak field . . . .'' [7] Therefore, the data demonstrate that perturbation theory is not valid in the plateau region or above cutoff. Recent experiments [8–20] have produced coherent radiation at wavelengths below 5 nm from both atoms and molecules for instantaneous optical intensities approaching $10^{18}\,\mathrm{W\,cm^{-2}}$. The availability of mode-locked Ti:sapphire lasers producing sub-30-fs pulses has resulted in the recent observation [16–20] of harmonics having photon energies of ≈500 eV. In addition to the harmonic spectra reported by several groups, data concerning the polarization characteristics [21–29], temporal coherence [30], and spatial properties [31] of the harmonic radiation have appeared in the literature.

Not only does high-order harmonic generation (HG) po-

 ©2000 The American Physical Society

tentially afford a table-top source of coherent XUV and soft-x-ray radiation, but it also offers a window onto the interaction between an atom and a strong optical field. Describing the harmonic response of the atom at high optical fields, and the presence and characteristics of the higher harmonics comprising the plateau and cutoff, in particular, requires nonperturbative theoretical methods, and several avenues have been explored [32–43]. By direct numerical integration of the time-dependent Schrödinger equation, Krause, Schafer, and Kulander [34–36] calculated the response of a single atom to a strong field, assuming the single electron approximation to be valid. However, the computational capacity required by this approach and the desirability for insight into the physical processes underlying HG has prompted the pursuit of other nonperturbative theoretical models. Several groups [28,33,38–43] have solved Schrödinger's equation by approximating the atomic potential, the number of bound atomic states, or ground-state depletion in different ways. The unifying theme in previous theoretical treatments of HG is the goal of solving for the dipole moment associated with an electron oscillating in an electric field produced by the fundamental optical field. In each instance, the interaction of the optical field with the electron is treated classically. Not surprisingly, then, existing calculations are consistent with the predictions of the semiclassical two-step model for HG which was proposed by Kulander *et al.* [36] and Corkum [37]. In this view, bound electrons tunnel into the continuum under the influence of the optical field, accelerate to a maximum energy of approximately three times the ponderomotive potential (U_p), and subsequently recombine with the parent ion to generate short-wavelength generation. Consequently, the function of the field is to suppress the Coulomb barrier and accelerate the electron. Harmonic radiation is produced only by scattering in which the electron approaches the ion along one of two primary trajectories having phase properties that vary quite differently with laser intensity. In order to conserve momentum and energy, this picture requires that photoelectrons are born in the continuum at rest. Weihe and Bucksbaum [25] have concisely summarized the key features of the two-step model and have pointed out that the various methods advanced within the framework of the model differ primarily in the manner by which the returning electron is assumed to be scattered by the core. To summarize, semiclassical theory assumes that harmonic radiation is produced primarily by "... (i) ... electrons that return to the nucleus, (ii) appear in the continuum with zero velocity, and (iii) ... have an appropriate kinetic energy to produce a given harmonic at the time of return." [38]. Notice that (ii) is at odds with experimental observations of above-threshold ionization, which is known to produce energy spectra comprising discrete peaks separated by integral multiples of the fundamental photon energy. Therefore, although the semiclassical model has proven quite successful in describing several facets of HG, for ATI and HG to be firmly linked will require that HG theory account for the discrete, nonzero energies of the photoelectrons produced by ATI.

An alternative framework for describing high-order harmonic generation, in which the optical field assumes a more active role in the process than is suggested by the semiclassical picture, was recently introduced [44]. Radiation is produced by transitions between quantized field ("quantum") Volkov states which are *stationary* solutions of Schrödinger's equation for a quantized optical field interacting with an electron. Once injected into the optical field by ATI, photoelectrons interact with photons to form Volkov states and the HG spectrum, for example, can be expressed in terms of the Volkov- to Volkov-state transition probability. Since quantum Volkov states are stationary, the total energy and momentum of the electron-photon system are well-defined, and formal scattering theory may be invoked; hence, such Volkov states may be regarded as the high optical field analog of bound states of the atom. The electron-photon interaction is dominant, which is to be expected for the optical intensities characteristic of HG experiments and is inevitable as peak intensities of the driving optical field continue to rise. Energy and momentum conservation, the generation of odd harmonics (rather than broadband radiation), and the spectral and polarization characteristics of each harmonic follow directly from the theory, which has no adjustable constants.

In this paper, the QED formalism for HG is described in detail. This treatment yields an analytic expression for the harmonic-generation matrix element that reproduces virtually all of the key experimental results in the literature but also presents a picture of the mechanisms responsible for HG that contrasts with the semiclassical view. Specifically, the discrete photoelectron energies produced by ATI are preserved throughout the HG process; that is, the electron maintains its momentum during Volkov- to Volkov-state transitions. Also, HG is an optical energy-conversion process in which q pump (fundamental) photons are annihilated to produce a single, harmonic ($q\hbar\omega$) photon. In this sense, it is related to nonlinear Compton scattering by which the conversion of as many as four fundamental photons into a single γ-ray photon has been observed [45]. The primary conclusion is that, although energy and momentum conservation require that the electron ultimately return to the ground state of the parent atom, the generation of harmonic radiation (including those harmonics having photon energies $q\hbar\omega$ comparable to or less than the atomic ionization potential) results from transitions between quantum Volkov states. Comparison of theoretical predictions with experimental harmonic spectra and polarization data for individual harmonics points to the conclusion that HG draws primarily upon only the lowest two or three photoelectron orders when the fundamental optical field duration is sufficiently long ($\gtrsim$100–150 fs) to allow electrons having energies of more than a few eV to escape from the parent atom during the laser pulse risetime. Recent experiments [16–20] in which the laser pulse width is $<$30 fs have, however, succeeded in effectively freezing the motion of more energetic ATI electrons during the pulse risetime, resulting in harmonic spectra comprising contributions from considerably more photoelectron orders. Because the higher ATI order (i.e., photoelectron energy) electrons are coupled more strongly to the photon field, the single atom response extends to higher-order harmonics than those accessible with long optical pulses. Consequently, the impact on HG spectra of capturing the higher energy

($\gtrsim$5–10 eV) photoelectrons is dramatic.

Before proceeding, a few comments regarding the suitability of applying QED theory to harmonic generation in intense optical fields are in order. The generally held view is that quantization effects vanish or are negligible in the limit of high photon number. However, it is well known that ATI and HG spectra consist of peaks separated by integral multiples of the fundamental photon energy, inherently reflecting the quantized nature of both processes. Said another way, a discrete number of photons interacts with a bound electron to produce an ATI electron or an HG photon, both having specific energies. While it is certainly true that the few photons necessary to photoionize an atom by ATI, for example, are negligible compared to the total number of photons in the field, it is argued here that accounting for the *incremental* number of photons absorbed from the optical field, annihilated to produce a harmonic photon, or released back to the field upon recombination, is essential for describing HG and its relationship to ATI. Treating the entire photon field and an electron as an isolated system allows one to provide a fully quantum description of HG in an intense field and offers a foundation on which similar phenomena at still higher intensities can be described.

This paper is organized in the following manner. Section II presents the derivation of the matrix element for the overall HG process, of which the critical component is the Volkov→Volkov-state transition matrix element. Comparison of the QED theoretical predictions with semiclassical results and experimental data is the subject of Sec. III, while Sec. IV summarizes the conclusions of this study and briefly discusses the implications of this QED framework for future work.

II. DERIVATION OF THE HARMONIC-GENERATION TRANSITION MATRIX

A. ATI and quantized field Volkov states

Keldysh-Faisal-Reiss (KFR) theory [46] provides a nonperturbative model for describing multiphoton ionization in terms of a transition of an electron from a bound atomic state to a classical Volkov state [47], a solution of the time-dependent Schrödinger's equation for a free electron oscillating in an electromagnetic field. Reiss [48] demonstrated that this semiclassical description of the ATI process reproduces the general characteristics of experimental spectra [49], but subsequent experiments and calculations by Mohideen *et al.* [50] have shown that the KFR matrix element does not accurately predict the ATI electron energy distribution for linearly or circularly polarized optical driving fields in the limit $U_p \gg E_b$, where U_p is the ponderomotive energy ($e^2\mathrm{E}^2/4m\omega^2 \approx 9.33\times10^{-14} I\lambda^2$, where E is the classical amplitude of the electric field, and I and λ are the laser intensity and wavelength expressed in W cm^{-2} and μm, respectively) and E_b is the ionization potential of the atom or molecule of interest.

In 1989, Guo *et al.* [51] reported the derivation of the matrix element for ATI in terms of formal scattering theory. If the electron-ion core Coulombic interaction is neglected when the electron is in the continuum, the matrix element can be expressed as

$$T_{fi} = \langle \Psi_f^- | V_1 | \Phi_i; l \rangle, \tag{1}$$

in which $|\Phi_i; l\rangle$ represents the initial state of the atom and photon field, where Φ_i is a bound atomic state that is decoupled from the photon state $|l\rangle$ of the fundamental optical field. The electron-optical field interaction is given by

$$V_1 = -\frac{e}{m}\vec{A}(\vec{k}\cdot\vec{r})\cdot(-i\vec{\nabla}) + \frac{e^2A^2(\vec{k}\cdot\vec{r})}{2m}, \tag{2}$$

where the vector potential is related to the phase of the photon field, Θ, by

$$\vec{A}(\vec{k}\cdot\vec{r}) = g(\hat{\varepsilon}e^{i\vec{k}\cdot\vec{r}}e^{i\Theta}\hat{a} + \hat{\varepsilon}^* e^{-i\vec{k}\cdot\vec{r}}e^{-i\Theta}\hat{a}^\dagger), \tag{3}$$

$g = (2V_\gamma \omega \varepsilon_0)^{-1/2}$ is the normalization factor arising from quantization of the field, V_γ is the normalization volume for the photon, ε_0 is the permittivity of free space, ω is the fundamental frequency, and $\hat{\varepsilon}$ is the polarization vector. In Eq. (1), the scattering state Ψ_f^- is a solution to the Lippmann-Schwinger equation [52]:

$$\Psi_f^- = [1 + G^- V_1]|\Phi_f; n\rangle, \tag{4}$$

in which the final state Φ_f is also decoupled from the optical field and Green's operator G^- is given by

$$G^- = \frac{1}{E - K - V_1 + i\alpha}, \tag{5}$$

where $K = (-i\vec{\nabla})^2/2m + N_a\omega$ is the noninteracting Hamiltonian for the electron and optical field, N_a is the number operator for the photon field, E is the total energy of the atom-photon field system, α is the spectral linewidth of the fundamental radiation, m is the rest mass of the electron, and natural units have been adopted (i.e., $\hbar \equiv c \equiv 1$). A nonperturbative expansion of Eq. (1) allows one to rewrite the ATI transition matrix element as

$$T_{fi} = \sum_{\vec{P}_1, n} \langle \Phi_f; n | \Psi_\mu \rangle \frac{i\alpha}{E - E_\mu + i\alpha} \langle \Psi_\mu | V_1 | \Phi_i; l \rangle, \tag{6}$$

where $\vec{P}_1$ is the electron momentum and the intermediate states Ψ_μ are known as quantized field (quantum) Volkov states [53], which are solutions of the equation

$$(K + V_1)|\Psi_\mu\rangle = E_\mu|\Psi_\mu\rangle = E_\mu|\vec{P}_1, n_1; 0\rangle, \tag{7}$$

and whose energy eigenvalues are given by

$$E_\mu = E(|\vec{P}_1, n_1; 0\rangle) = \frac{|\vec{P}_1|^2}{2m} + n_1\omega + \frac{e^2g^2n_1}{m}, \tag{8}$$

which, in the large photon number limit, becomes

$$E_\mu = \frac{|\vec{P}_1|^2}{2m} + n_1\omega + z\omega, \tag{9}$$

where z is the ponderomotive energy (U_p) normalized to the fundamental photon energy and, since the intensity I can be written

$$I = \frac{n_1}{V_\gamma}\omega, \tag{10}$$

then

$$U_p = z\omega = \frac{e^2 g^2 n_1}{m} = \frac{e^2 I}{2m\varepsilon_0\omega^2}. \tag{11}$$

As noted earlier, the Ψ_μ are stationary solutions to Schrödinger's equation that are a direct consequence of quantizing the optical field and treating the electron and photons as an isolated system. Consequently, the energy and momentum of the system are well-defined. Notice that the Volkov-state eigenenergies depend on both U_p and the ATI electron energy and are separated by integral multiples of the fundamental field photon energy.

Prior to the work of Guo *et al.* [51,53,54], KFR theory of ATI addressed only the last term on the right-hand side of Eq. (6) and assumed the final state(s) of the scattering process to be classical Volkov states. Although this approach explains several general characteristics of ATI, it neglects the fact that ATI experiments collect electrons outside the laser-electron interaction region, thus rendering the Volkov state an inappropriate choice for the final state of the process. It is therefore necessary to include a matrix element that describes transitions from quantized Volkov states to ''field-free'' (i.e., decoupled from the field) states of the electron. Specifically, Guo *et al.* [54,55] demonstrate that the final state for the ATI process is the product of free electron and free photon field states and that a transition from the quantum Volkov state to this final state must be included in the description of ATI. By extending KFR theory to quantize the electron-photon interaction, the treatment of Refs. [54] and [55] successfully reproduces experiments [56] in which the Kapitza-Dirac effect was observed in an intense laser field (10^{13}–$10^{14}\,\mathrm{W\,cm^{-2}}$). In these calculations, it is the Volkov-to-free-state transition that accounts for the primary features in the angular distribution of the scattered ATI photoelectrons.

B. Development of the harmonic-generation transition matrix: General considerations

Although Eq. (6) provides an accurate description of ATI, it includes only a single (fundamental) radiation field and, therefore, offers no mechanism for generating harmonics. In order to account for the interaction of the fundamental with a secondary photon field, the transition matrix T_{fi} will be modified to include a perturbation term. Since the intensities of the harmonics are typically small relative to that of the fundamental, time-independent perturbation theory may be invoked to calculate the modified transition matrix and perturbed Volkov-state wave function.

The introduction of the secondary optical field gives rise to the interaction term

$$V' = -\frac{e}{m}\vec{A}'(\vec{k}'\cdot\vec{r})\cdot(-i\vec{\nabla}) + \frac{e^2}{m}\vec{A}(\vec{k}\cdot\vec{r})\cdot\vec{A}'(\vec{k}'\cdot\vec{r}), \tag{12}$$

in which

$$\vec{A}'(\vec{k}'\cdot\vec{r}) = g'(\hat{\varepsilon}' e^{i\vec{k}'\cdot\vec{r}}\hat{a}' + \hat{\varepsilon}'^{*} e^{-i\vec{k}'\cdot\vec{r}}\hat{a}'^{\dagger}) \tag{13}$$

and primes denote quantities associated with harmonics. Consequently, the unperturbed and perturbed Hamiltonians for the system can be written, respectively, as

$$H_0 = K + V_1 \tag{14}$$

and

$$H = K + V_2 + V', \tag{15}$$

where V_1 and V_2 are given by the expression

$$V_s = -\frac{e}{m}\vec{P}\cdot\vec{A}(\Theta_s) + \frac{e^2}{2m}\vec{A}^2(\Theta_s), \quad s = 1,2. \tag{16}$$

Notice that the perturbed Hamiltonian includes both the secondary interaction V' and V_2. The latter differs from V_1 only by the shift in the phase of the fundamental field, $\Delta\Theta = \Theta_2 - \Theta_1$, which is a direct consequence of the electron-photon field scattering process and the concomitant generation of the harmonic field.

Consequently, Schrödinger's equation for the perturbed Hamiltonian H can now be written as

$$[K + V_2 + V'](\Psi_\mu + \Psi'_\mu) = E_\mu(\Psi_\mu + \Psi'_\mu), \tag{17}$$

in which Ψ_μ, the quantized field Volkov states, are the eigenfunctions associated with the Hamiltonian $K + V_1$ [Eq. (7)], the eigenenergies E_μ are defined in Eqs. (7)–(9), and Ψ'_μ is the perturbed Volkov-state wave function. By neglecting the term $V'\Psi'_\mu$ in Eq. (17) and invoking Eq. (7), the perturbed Volkov state can be expressed as

$$\Psi'_\mu = -\sum_\nu \frac{|\Psi_\nu\rangle\langle\Psi_\nu|(V_2 - V_1) + V'|\Psi_\mu\rangle}{E_\nu - E_\mu} = -\sum_\nu \frac{|\Psi_\nu\rangle\langle\Psi_\nu|(E_\nu - E_\mu) + V'|\Psi_\mu\rangle}{E_\nu - E_\mu}, \tag{18}$$

where $|\Psi_\nu\rangle$ are the quantum Volkov state solutions for the equation:

$$(K + V_2)|\Psi_\nu\rangle = E_\nu|\Psi_\nu\rangle = E_\nu|\vec{P}_2, n_2; 1\rangle \tag{19}$$

and whose energy eigenvalues are given by

$$E_\nu = E(|\vec{P}_2, n_2; 1\rangle) = \frac{|\vec{P}_2|^2}{2m} + n_2\omega + \frac{e^2 g^2 n_2}{m} + \omega'. \tag{20}$$

Evaluated at resonance ($E_\nu \rightarrow E_\mu$), the matrix element associated with $(V_2 - V_1)$ vanishes and Eq. (18) becomes

$$\Psi'_\mu = -\sum_\nu \frac{|\Psi_\nu\rangle\langle\Psi_\nu|V'|\Psi_\mu\rangle}{E_\nu - E_\mu}, \tag{21}$$

which is identical to Eq. (2) of Ref. [44]. By substituting $\Psi_\mu + \Psi'_\mu$ for Ψ_μ, equation (6) can now be modified to incorporate the secondary field interaction. Of the resulting terms contributing to harmonic generation, the dominant matrix element is:

$$\begin{aligned} T_{fi} &= \sum_\mu \langle \Phi_f; m; 1|\Psi'_\mu\rangle \frac{i\alpha}{E - E_\mu + i\alpha}\langle \Psi_\mu|V_1|\Phi_i; l; 0\rangle \\ &= \sum_{(\vec{P}_1, n_1)(\vec{P}_2, n_2)} \langle \Phi_f; m; 1|\vec{P}_2, n_2; 1\rangle \\ &\quad \times \langle \vec{P}_2, n_2; 1|V'|\vec{P}_1, n_1; 0\rangle \\ &\quad \times \frac{1}{E(\vec{P}_1, n_1; 0) - E(\vec{P}_2, n_2; 1)}\langle \vec{P}_1, n_1; 0|V_1|\Phi_i; l; 0\rangle \\ &\quad \times \frac{i\alpha}{E - E(\vec{P}_1, n_1; 0) + i\alpha}. \end{aligned} \tag{22}$$

At resonance, the two linewidth factors in Eq. (22) yield expressions for the relationships between the initial, intermediate, and final states of the atom-photon field system:

$$E(\vec{P}_1, n_1; 0) = \frac{|\vec{P}_1|^2}{2m} + n_1\omega + z\omega = E \tag{23}$$

and

$$E(\vec{P}_2, n_2; 1) = \frac{|\vec{P}_2|^2}{2m} + n_2\omega + z\omega + \omega' = E(\vec{P}_1, n_1; 0), \tag{24}$$

where

$$E = -E_b + l\omega = -E_b + m\omega + \omega', \tag{25}$$

and ω and ω' are the fundamental and harmonic frequencies, respectively. Substituting Eq. (25) into Eq. (23) results in the well-known relationship for the energies of ATI photoelectrons:

$$\frac{|\vec{P}_1|^2}{2m} = (l - n_1)\omega - E_b - z\omega \equiv j\omega - E_b - z\omega, \tag{26}$$

where j is an integer representing the number of photons absorbed in the ATI process. When combined with the expression for momentum conservation, Eq. (23) also leads directly to the harmonic condition which will be presented in the next section. Evaluated at resonance ($E - E(\vec{P}_1, n_1; 0) \rightarrow 0$ and $E(\vec{P}_2, n_2; 1) - E(\vec{P}_1, n_1; 0) \rightarrow 0$), Eq. (22) yields the transition matrix for harmonic generation, denoted T_{fi}^{HG} and given by

$$\begin{aligned} T_{fi}^{\mathrm{HG}} &= \sum_{(\vec{P}_1, n_1)(\vec{P}_2, n_2)} \langle \Phi_f; m; 1|\vec{P}_2, n_2; 1\rangle \\ &\quad \times \langle \vec{P}_2, n_2; 1|V'|\vec{P}_1, n_1; 0\rangle\langle \vec{P}_1, n_1; 0|V_1|\Phi_i; l; 0\rangle. \end{aligned} \tag{27}$$

The derivation of an analytical expression for T_{fi}^{HG} is the subject of the next section. Notice, however, that in Eqs. (22) and (27), one photon now appears in the secondary photon field. Also, recall that the final state Φ_f for an ionization process such as ATI is that for a free electron and is often approximated by a plane wave. As will be discussed in the next section, conserving momentum and energy in the harmonic-generation process requires that Φ_f be the initial, bound state of the electron Φ_i. Consequently, the net effect of harmonic generation is to convert q photons at the fundamental into one photon at the qth harmonic.

Before proceeding, several comments regarding Eq. (27) should be made. The picture presented by this equation differs markedly from the semiclassical view in that the optical field is no longer an external agent. Rather, the electron and photon field interact through the Hamiltonian V_1 to form an entity having stationary states. Borrowing a term from physical chemistry, we refer to the transient electron-photon species as a ''complex.'' Let us now consider the second term on the right-hand side of Eq. (27), in particular. As discussed in the next section, this term describes transitions between quantum Volkov states in which the interaction term V' is attributable to the harmonic radiation field. As we shall see, this approach yields an analytic expression for the harmonic-generation rate that is in accord with experiment but requires that the fundamental optical field undergo a phase shift during Volkov- to Volkov-state transitions. Such a phase shift is precisely what one would expect for the situation in which an optical field and an electron scatter to produce a new (harmonic) radiation field. As shown briefly in the Appendix, a change in the phase of the fundamental optical field is required by the continuity condition and energy conservation at the scattering plane.

Notice also that this theory does not invoke the dipole approximation. Rather than adopting the dipole gauge $(\vec{E}\cdot\vec{r})$, the electron-optical field interaction is described by the minimum coupling expression:

$$\frac{(\vec{P} - e\vec{A} - e\vec{A}')^2}{2m} = \cdots - \frac{e\vec{P}\cdot\vec{A}}{m} - \frac{e\vec{P}\cdot\vec{A}'}{m} + \frac{e^2}{2m}\vec{A}\cdot\vec{A}'. \tag{28}$$

Normally, the last term on the right-hand side of Eq. (28) is dropped. A key aspect of the theory presented here (and in Refs. [44] and [73]) is that, because of the magnitude of the optical field intensities, retention of the $\vec{A}\cdot\vec{A}'$ term is now essential. Indeed, this second-order term is key to describing the interaction of the fundamental and harmonic radiation

fields [cf. Eq. (12)] and, specifically, the conversion of fundamental photons into harmonic radiation. As discussed earlier, harmonic photons in this framework are produced by transitions between stationary states of the e^--photon system rather than by decelerating an electron.

C. Calculation of T_{fi}^{HG}

The transition matrix for the harmonic-generation process, T_{fi}^{HG}, is composed of three matrix elements, all of which explicitly involve the nonrelativistic quantized field Volkov state [53,57,58],

$$|\Psi_\mu\rangle=|\vec{P}_s,n_s\rangle=V_e^{-1/2}e^{i(\vec{P}_s+z\vec{k})\cdot\vec{r}}e^{-iN_d\vec{k}\cdot\vec{r}}e^{-F_s}|n_s\rangle, \tag{29}$$

in which the subscript s denotes a specific Volkov state (denoted 1, 2, . . .), V_e is the normalization volume of the electron, and the coupling between the electron and optical field is represented by the operator $\exp(-F_s)$, where

$$F_s=\frac{eg}{m\omega}(\vec{P}_s\cdot\hat{\varepsilon}e^{i\Theta_s}\hat{a}-\vec{P}_s\cdot\hat{\varepsilon}^*e^{-i\Theta_s}\hat{a}^\dagger)+\frac{e^2g^2\cos\xi}{4m\omega}(e^{-i2\Theta_s}\hat{a}^{\dagger 2}-e^{i2\Theta_s}\hat{a}^2), \tag{30}$$

$\hat{\varepsilon}$ is the polarization vector, and ξ expresses the degree of polarization of the optical field ($\xi=0$ corresponds to linear polarization, $\xi=\pi/2$ to circular polarization). Specifically, for a radiation field of arbitrary polarization, the polarization vector is given by

$$\hat{\varepsilon}=[\hat{\varepsilon}_x\cos(\xi/2)+i\hat{\varepsilon}_y\sin(\xi/2)], \tag{31}$$

where ξ is related to the ellipticity μ by

$$\mu=\tan\frac{\xi}{2}. \tag{32}$$

The third term on the right-hand side of Eq. (27), $\langle\vec{P}_1,n_1;0|V_1|\Phi_i;l;0\rangle$, represents the multiphoton ionization of an atom from bound state Φ_i to a quantum Volkov state:

$$T^{\mathrm{KFR}}\equiv\langle\vec{P}_1,n_1;0|V_1|\Phi_i;l;0\rangle=\omega(z-j)\mathcal{J}_{-j}(\zeta,\eta,\phi)\Phi_i(\vec{P}_1+z\vec{k}-j\vec{k}), \tag{33}$$

where $\Phi_1(\vec{P}_1+z\vec{k}-j\vec{k})$ is the momentum-space representation (Fourier transform) of the bound atomic state wave function Φ_i, and $\mathcal{J}$ is the generalized Bessel function, defined by

$$\mathcal{J}_j(x,y,\phi)=\sum_{m=-\infty}^{\infty}J_{-j-2m}(x)J_m(y)e^{i2m\phi}, \tag{34}$$

where J is the ordinary Bessel function. The arguments of $\mathcal{J}_{-j}$ in Eq. (33) are

$$\zeta=\left|\frac{2e\mathrm{E}}{m\omega^2}\vec{P}_1\cdot\hat{\varepsilon}\right|, \tag{35}$$

$$\eta=-\frac{z}{2}\cos\xi, \tag{36}$$

$$\phi=\arg(\vec{P}_1\cdot\hat{\varepsilon}). \tag{37}$$

Note that incorporating the classical field amplitude E into ζ is valid in the limit of large photon number (i.e., $g\omega\sqrt{n}\rightarrow\mathrm{E}$). The notation KFR in Eq. (33) reflects the fact that, in the large photon number approximation, this matrix element is identical to that in KFR theory [46]. Also, recall that the magnitude of $\vec{P}_1$ takes on only the discrete values [Eq. (26)]

$$|\vec{P}_1|=P_j=\sqrt{2m\omega(j-z-E_b/\omega)}. \tag{38}$$

Therefore, $\zeta_j=(2e\mathrm{E}/m\omega^2)P_j\cos\varphi$, where φ is the angle between $\vec{P}$ and $\hat{\varepsilon}$. As noted earlier, predictions of ATI spectra on the basis of Eq. (33), in which the electron kinetic energy is given by Eq. (38), are in general agreement with experiment [49]. Of course, the final state in the ATI process is not a Volkov state and the additional factor [not present in Eq. (33)] that describes the transition to a free-electron state does modify the overall profile of the ATI spectrum but *does not* affect the energies of the photoelectrons [55].

The first term on the right-hand side of Eq. (27), $\langle\Phi_i=\Phi_f;m;1|\vec{P}_2,n_2;1\rangle$, describes the transition from a quantum Volkov state back to the ground state of the atom and is given by

$$\langle\Phi_i;m;1|\vec{P}_2,n_2;1\rangle=\frac{(2\pi)^3}{V_e}\Phi_i^*(\vec{P}_2+z\vec{k}-j'\vec{k})\mathcal{J}_{-j'}^*(\zeta,\eta,\phi), \tag{39}$$

where $j'=m-n_2$ is the number of photons that revert to the pump field during this transition.

The central theme of this paper is that the middle term of Eq. (27), denoted T^V, is the matrix element that describes the essence of the HG process and is instrumental in calculating the spectrum and other physical characteristics associated with the process, including the polarization of individual harmonics. It must also be emphasized that T^V has no classical analog. This matrix element represents transitions between quantum Volkov states and can be expanded as

$$T^V=\langle\vec{P}_2,n_2;1|V'|\vec{P}_1,n_1;0\rangle=\left(\frac{e^2g'}{m}\right)\frac{(2\pi)^3}{V_e}\delta(\vec{P}_2+n_2\vec{k}+\vec{k}',\vec{P}_1+n_1\vec{k})\times\langle n_2|e^{F_2}[\vec{P}_1-eg(\hat{\varepsilon}\hat{a}+\hat{\varepsilon}^*\hat{a}^\dagger)]\cdot\hat{\varepsilon}'^*e^{-F_1}|n_1\rangle, \tag{40}$$

where the δ function enforces momentum conservation during the Volkov→Volkov transition. That is,

$$\vec{P}_2+n_2\vec{k}+\vec{k}'=\vec{P}_1+n_1\vec{k}. \tag{41}$$

Combining Eq. (41) with the resonance condition [Eq. (24)] yields expressions defining the mode $(\omega',\vec{k}')$ of the secondary field:

$$\omega' = (n_1 - n_2)\omega \equiv q\omega, \tag{42}$$

$$\vec{k}' = (n_1 - n_2)\vec{k} \equiv q\vec{k}, \tag{43}$$

where q is the harmonic order and, therefore,

$$\vec{P}_2 = \vec{P}_1 \equiv \vec{P}. \tag{44}$$

We conclude that the momentum of the photoelectron (defined relative to the parent atom) is unchanged by Volkov→Volkov-state transitions. Also, a comparison of Eqs. (25) and (42) shows that $l-m=n_1-n_2=q$ and, consequently, $j=j'$. These relationships confirm that harmonic generation is a direct conversion process in which q fundamental photons $(\omega,\vec{k})$ are annihilated by the electron-photon complex to produce one harmonic photon $(\omega',\vec{k}')$, followed by the eventual return of the electron to the original bound state, Φ_i. Conversely, if the final atomic state Φ_f is *not* Φ_i, then $\omega' \neq (l-m)\omega$ and the radiation generated cannot be a harmonic of the fundamental field. For this reason, the photoelectrons detected in ATI experiments, for example, are not responsible for harmonic generation for they carry away momentum and energy that cannot be balanced by harmonic radiation.

The critical step in calculating Eq. (40) is evaluating the matrix elements $\langle n_2|e^{F_2-F_1}|n_1\rangle$, $\langle n_2|e^{F_2}\hat{a}e^{-F_1}|n_1\rangle$, and $\langle n_2|e^{F_2}\hat{a}^\dagger e^{-F_1}|n_1\rangle$. Algebraic manipulation leads to the expressions

$$\langle n_2|e^{F_2-F_1|n_1}\rangle = e^{iq[(\Delta\Theta/2)+\phi]}\mathcal{J}_{-q}\left(x,y,-\frac{\pi}{4}-\phi\right) \tag{45}$$

and

$$\begin{aligned}\langle n_2|e^{F_2}\hat{a}e^{-F_1}|n_1\rangle &= \sqrt{n_1}\langle n_2|e^{F_2-F_1}|n_1-1\rangle \\ &= e^{i(q-1)(\Delta\Theta/2+\phi)}\sqrt{n_1}\,\mathcal{J}_{-q+1}\left(x,y,-\frac{\pi}{4}-\phi\right),\end{aligned}$$

$$\begin{aligned}\langle n_2|e^{F_2}\hat{a}^\dagger e^{-F_1}|n_1\rangle &= \sqrt{n_1+1}\langle n_2|e^{F_2-F_1}|n_1+1\rangle \\ &= e^{i(q+1)(\Delta\Theta/2+\phi)}\sqrt{n_1+1} \\ &\quad\times\mathcal{J}_{-q-1}\left(x,y,-\frac{\pi}{4}-\phi\right),\end{aligned} \tag{46}$$

where, in the large photon-number approximation,

$$x = \left|\frac{4e\mathrm{E}}{m\omega^2}\vec{P}_1\cdot\hat{\varepsilon}\right|\sin\frac{\Delta\Theta}{2} = 2\zeta\sin\frac{\Delta\Theta}{2}, \tag{47}$$

$$y = -z\cos\xi\sin\Delta\Theta = 2\eta\sin\Delta\Theta, \tag{48}$$

and $\Delta\Theta=\Theta_2-\Theta_1$ is the shift in the phase of the fundamental optical field that occurs during transitions between the quantum Volkov states $|\vec{P}_1,n_1;0\rangle$ and $|\vec{P}_2,n_2;1\rangle$.

The dependence of the matrix element for Volkov→Volkov-state transitions on $\Delta\Theta$ is a key aspect of this theory and is reminiscent of electron scattering from a potential in which a phase shift occurs that determines the scattering cross section [59]. Specifically, the present calculations suggest that generating harmonics *requires* the existence of a shift in the phase of the fundamental field during transitions between quantum Volkov states. The implications of the requirement imposed by the electron-photon field interaction that $\Delta\Theta$ be nonzero are discussed in the Appendix.

Evaluating T_{fi}^{HG} [Eq. (27)] requires summing the contributions from all ATI processes (i.e., integrating over $\vec{P}$ and summing over n). In doing so, the summation over n_2 [Eq. (40)] is replaced by summing over q since $q=n_1-n_2$. With these considerations in mind, the transition matrix for the harmonic-generation process T_{fi}^{HG} becomes, subject to the harmonic condition [Eqs. (42) and (43)],

$$\begin{aligned}T_{fi}^{\mathrm{HG}} = \sum_q \sum_{\vec{P},n_1} &\frac{(2\pi)^3}{V_e}\left(\frac{g'\omega^2}{2\mathrm{E}}\right)\omega(z-j)\left|\mathcal{J}_{-j}(\zeta,\eta,\phi)\right|^2 \\ &\times\left|\Phi_i(\vec{P}+z\vec{k}-j\vec{k})\right|^2 e^{iq[(\Delta\Theta/2)+\phi]} \\ &\times\left\{\frac{2e^2\mathrm{E}^2}{m\omega^3}e^{-i[(\Delta\Theta/2)+\phi]}\mathcal{J}_{-q+1}(x,y,-\phi-\pi/4)\hat{\varepsilon}\cdot\hat{\varepsilon}'^*\right. \\ &+\frac{2e^2\mathrm{E}^2}{m\omega^3}e^{i[(\Delta\Theta/2)+\phi]}\mathcal{J}_{-q-1}(x,y,-\phi-\pi/4)\hat{\varepsilon}^*\cdot\hat{\varepsilon}'^* \\ &\left.-\frac{2e\mathrm{E}}{m\omega^2}\vec{P}\cdot\hat{\varepsilon}'^*\mathcal{J}_{-q}(x,y,-\phi-\pi/4)\right\},\end{aligned} \tag{49}$$

where $\vec{P}_1=\vec{P}_2\equiv\vec{P}$ is the photoelectron momentum inside the laser (fundamental) field. Recalling now that $n_1=l-j$, then the summation over n_1 can, for a fixed initial photon number l, be replaced by a summation over j. Summing over $\vec{P}$ is equivalent to integrating over the polar angles θ and ψ in a spherical coordinate system with $|\vec{P}|$ constrained by Eq. (38). That is,

$$\sum_{\vec{P}} \rightarrow \int d\vec{P}^3 = \frac{V_e}{(2\pi)^3}\int_0^{2\pi}d\psi\int_0^{\pi}\sin\theta\,d\theta\int_0^{\infty}dP\cdot|\vec{P}|^2, \tag{50}$$

in which the azimuthal axis is chosen to coincide with $\vec{k}$ (i.e., $\vec{k}\cdot\vec{P}\propto\cos\theta$). Consequently, the transition matrix for all harmonics generated can be rewritten for a specified harmonic order q [from Eq. (49)] as

$$\begin{aligned}T_q^{\mathrm{HG}} = e^{iq[(\Delta\Theta/2)+\phi]}\sum_j &\left(\frac{g'\omega^2}{2\mathrm{E}}\right) \\ &\times(z-j)\sqrt{2m^3\omega^5(j-z-E_b)}\times\int_0^{2\pi}d\psi \\ &\times\int_0^{\pi}\sin\theta d\theta\left|\mathcal{J}_{-j}(\zeta_j,\eta,\phi)\right|^2\left|\Phi_i(\vec{P}_j+z\vec{k}-j\vec{k})\right|^2\end{aligned}$$

$$\times\left\{-\frac{2e\mathrm{E}}{m\omega^2}\vec{P}_j\cdot\hat{\varepsilon}'^*\mathcal{J}_{-q}(x_j,y,-\phi-\pi/4)\right.$$
$$+\frac{2e^2\mathrm{E}^2}{m\omega^3}e^{-i[(\Delta\Theta/2)+\phi]}\mathcal{J}_{-q+1}(x_j,y,-\phi-\pi/4)\hat{\varepsilon}\cdot\hat{\varepsilon}'^*$$
$$+\frac{2e^2\mathrm{E}^2}{m\omega^3}e^{i[(\Delta\Theta/2)+\phi]}\mathcal{J}_{-q-1}(x_j,y,-\phi$$
$$\left.-\pi/4)\hat{\varepsilon}^*\cdot\hat{\varepsilon}'^*\right\}, \tag{51}$$

in which the subscript j has been appended to the photoelectron momentum, ζ, and x throughout the equation to emphasize the quantized nature of the magnitude of $\vec{P}$ and, therefore, the discrete energies of the electrons.

The generation rate for a particular harmonic is proportional to $|T_q^{\mathrm{HG}}|^2$. That is,

$$\frac{dW_q(\Delta\Theta)}{d^3\vec{k}'}=2\pi\delta(\omega',q\omega)|T_q^{\mathrm{HG}}|^2, \tag{52}$$

where $dW_q/d^3\vec{k}'$ is the differential harmonic-generation rate. Since it was shown earlier that the harmonics are generated along the $\vec{k}$ direction of the fundamental field ($\vec{k}'=q\vec{k}$), the total harmonic-generation rate is given by

$$W_q(\Delta\Theta)=\frac{V'_\gamma}{(2\pi)^3}\int d\Omega_{\vec{k}'},2\pi|T_q^{\mathrm{HG}}|^2=\frac{V'_\gamma}{2\pi^2}|T_q^{\mathrm{HG}}|^2. \tag{53}$$

In calculating T_q^{HG}, the contributions of the photoelectron orders to the momentum $\vec{P}$ should be summed coherently. Because $\mathcal{J}(x,y,\phi)$ is an even or odd function of $\vec{P}$ (depending upon the order q),

$$\mathcal{J}_{-q}(x,y,\phi)=(-1)^{-q}\mathcal{J}_{-q}(-x,y,\phi), \tag{54}$$

where $x\propto\vec{P}\cdot\hat{\varepsilon}$, then integrating over the direction of $\vec{P}$ eliminates the even harmonics. However, the harmonic-generation rate is also dependent upon the phase shift $\Delta\Theta$, which must be summed incoherently because each value of $\Delta\Theta$ corresponds to an independent exit channel for the scattering process. Consequently, rewriting Eq. (53) to incorporate an integration over $\Delta\Theta$ yields

$$W_q=\frac{V'_\gamma}{2\pi^2}\int_0^{2\pi}d(\Delta\Theta)|T_q^{\mathrm{HG}}|^2. \tag{55}$$

From the analytic expressions for the transition matrix T_q^{HG} and harmonic-generation rate W_q [Eqs. (51) and (55), respectively], one can calculate the harmonic spectrum (intensity as a function of harmonic order q) or the polarization characteristics, for example, of a specific harmonic (i.e., fixed q). Notice that the spectrum is expressed in terms of the transition probability associated with pairs of Volkov states. Comparison of calculations with experiments demonstrates that it is T^V [Eq. (40)] that is primarily responsible for harmonic generation and the properties of individual harmonics. It should be reiterated that this theory includes no adjustable constants and the parameters required for calculations are the laser intensity and wavelength, polarization, the ground-state wave function for the atom or molecule of interest, and the value(s) of j, the number of absorbed photons. The significance of the latter will be discussed in more detail in Sec. III. Furthermore, the discrete nature of the photoelectron energies, the harmonic condition, and the generation of only odd harmonics follow directly from the theory. Momentum conservation during Volkov- to Volkov state transitions predicts the highly directional nature of the harmonic radiation observed experimentally and the transition matrix has been derived for a fundamental optical field of arbitrary polarization. Finally, although propagation effects are known to be important in a variety of experimental situations, only the single atom response and the implications of QED theory for interpreting experimental results are considered here.

D. Discussion of the harmonic-generation transition matrix

Before proceeding further and comparing the analytic expression of Eq. (51) with experimental and semiclassical theoretical results, let us consider several implications of the theory presented in the preceding section. The initial step in the HG process, ATI, is represented by T^{KFR} [Eq. (33)], which describes the multiphoton ionization process in which j fundamental photons are absorbed. The initial number of photons l is reduced to $n_1=l-j$ as the electron undergoes a transition from the bound state Φ_i to a quantized field Volkov state, acquiring momentum $\vec{P}$. As discussed previously, the electron cannot, at the terminus of this transition, be regarded as being free. Rather, ATI can be viewed as the mechanism by which electrons of discrete energies are injected into the optical field.

The matrix element T^V that describes transitions between quantum Volkov states has several prominent characteristics: (i) each Volkov→Volkov-state transition generates an additional photon in the harmonic field, (ii) with each transition, the fundamental optical field loses $q=n_1-n_2$ photons, (iii) the electron maintains its momentum during the transition, and (iv) a shift in the phase of the fundamental field is required. Also, as demonstrated by Eq. (51), the harmonic-generation rate depends upon both the laser intensity and electron momentum. The centrality of Volkov- to Volkov-transitions to the theory proposed here stands in contrast to the semiclassical view in which harmonic radiation is produced only when an electron reencounters the parent atom and continuum-continuum transitions are considered to be negligible [38,60].

The generation of photons by transitions between stationary states of the electron-atom system undergirds atomic (and molecular) spectroscopy and it is proposed that an analogous situation occurs at high optical intensities except that the focus now shifts to the electron-photon system. In this sense, QED theory and Volkov- to Volkov-state transitions, in particular, can be viewed as the strong optical field analog of the stationary states of the electron-atom system in which the interaction term arises from the Coulombic field.

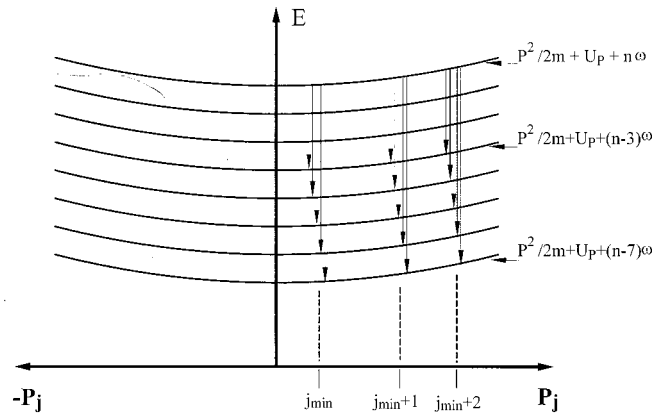


FIG. 1. Energy-level diagram qualitatively illustrating transitions between quantum Volkov states for several discrete photoelectron energies (first through third photoelectron orders). The minimum number of photons necessary to ionize the atom is denoted $j_{\min}$.

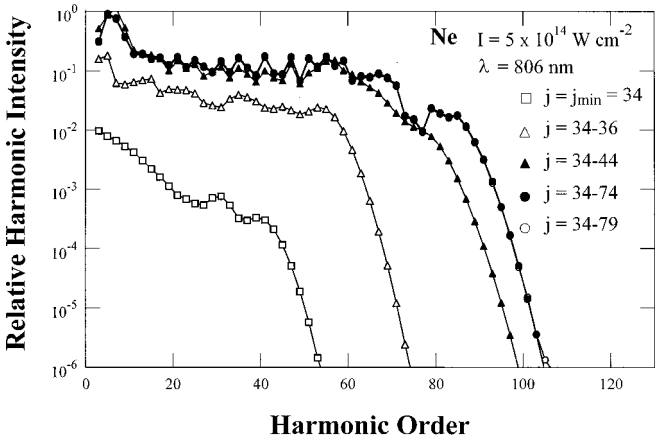


FIG. 2. Harmonic spectra calculated from Eq. (55) for Ne, assuming an intensity of $5\times10^{14}\,\mathrm{W\,cm^{-2}}$ and a fundamental wavelength of 806 nm. Results are shown for several sets of photoelectron orders: $j=j_{\min}=34$ (□), $j=34-36$ (△), $j=34-44$ (▲), $j=34-74$ (●), and $j=34-79$ (○). Convergence occurs rapidly when more than approximately five photoelectron orders are included in the calculations.

The manner in which the QED mechanism for generating harmonics is intertwined with the ATI process is qualitatively illustrated in Fig. 1, which shows the dependence of E on electron momentum $\vec{P}$. For each photoelectron order, transitions between quantum Volkov states yield photons that are multiples of the fundamental photon energy. This scenario preserves the discrete energies of photoelectrons produced by multiphoton ionization, and offers an explanation for harmonic orders having photon energies well above, comparable to, or less than, E_b. Note, too, that the theory permits more than one Volkov→Volkov transition to occur, in a manner analogous to radiative cascade in an atom, prior to recombination of the electron with the parent atom. Another interesting aspect of this theoretical treatment is that, as one may recall, the elimination of even harmonics results from integrating over the direction of $\vec{P}$. If the symmetry of the system is broken, this restriction would no longer be valid and all harmonics would emerge from the interaction region. This observation is interesting in light of the report by Englert and Rinehart [61] of the generation of the second harmonic of the Nd:YAG laser (1.06 μm) when the fundamental, having a peak intensity of $\sim10^{14}\,\mathrm{W\,cm^{-2}}$, interacted with a beam of 500-eV electrons. Similar experiments by Bula *et al.*[45] with relativistic electrons having energies of ~42 GeV yielded strong evidence for nonlinear Compton scattering $(e+n\omega_0\rightarrow\omega+e')$, where $2\leqslant n\leqslant 4$.

The final transition of the HG process entails the return of the electron to its original (bound) state and the release of the ponderomotive energy. Also, j photons are returned to the fundamental optical field. This process is essentially the reverse of ATI with the difference that q fundamental photons have already been converted into a harmonic photon.

III. COMPARISON OF QED THEORY WITH SEMICLASSICAL THEORY AND EXPERIMENT

A. General considerations: Cutoff in harmonic spectra and the role of *j*

In calculating harmonic spectra from Eq. (55), several issues warrant further examination. Since Eq. (51) involves a summation over j, it is appropriate to first verify that the series converges. Representative results shown in Fig. 2 confirm that convergence does, indeed, occur. Harmonic spectra are presented in the figure for Ne and several sets of photoelectron orders, assuming an intensity of $5\times10^{14}\,\mathrm{W\,cm^{-2}}$ and a laser wavelength of 806 nm. For these parameters, $U_p\approx30$ eV, the minimum number of absorbed photons necessary to ionize the atom (denoted $j_{\min}$) is 34, and convergence occurs rapidly when more than the first several photoelectron orders (defined as $j-j_{\min}+1$) are incorporated into the calculations. Notice that convergence first occurs in the plateau and, for the experimental parameters of Fig. 2, the spectra are indistinguishable once fewer than 20 ATI orders are included.

It is also evident from Fig. 2 (and Fig. 1 of Ref. [44]) that including successively larger values of j in the calculations extends the harmonic response to higher orders and, hence, shorter wavelengths. All of the spectra reproduce the plateau and cutoff that are observed experimentally and, if the laser intensity is held constant, cutoff moves to shorter wavelengths as the kinetic energy of the electrons is increased. This is a direct result of the fact that the generalized Bessel functions reflect the coupling between the electron and optical field through the interaction terms

$$-\frac{e}{m}(-i\vec{\nabla})\cdot\vec{A}(\vec{k}\cdot\vec{r})\propto\left(\frac{P^2z}{2m}\right)^{1/2} \tag{56}$$

and

$$\frac{e^2A^2(\vec{k}\cdot\vec{r})}{2m}\propto z. \tag{57}$$

Consequently, for a fixed laser intensity, the higher-order photoelectrons are coupled more strongly to the photon field, resulting in a single atom response extending to higher har-

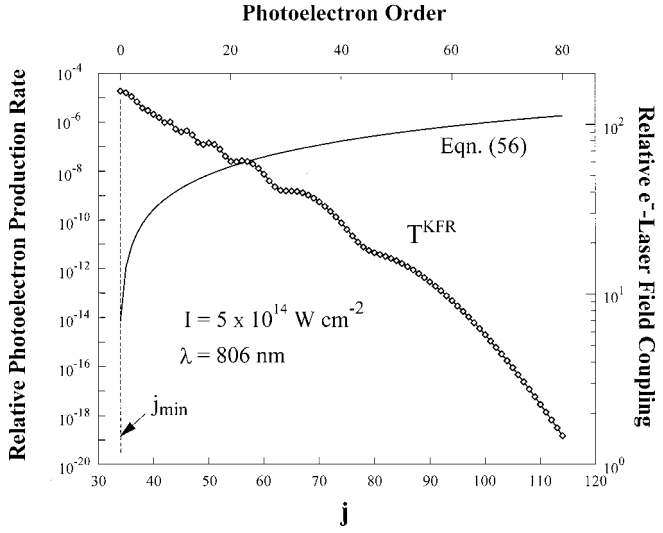


FIG. 3. Dependence of T^{KFR} [Eq. (33)] and one of two electron-photon field coupling terms [Eq. (56)] on j and the photoelectron order $(j-j_{\min}+1)$. The calculations assume a laser wavelength and intensity of 806 nm and $5\times10^{14}\,\mathrm{W\,cm^{-2}}$, respectively.

monics. If a means can be devised, therefore, to trap the higher-energy electrons, one would expect that shorter wavelengths will be generated more efficiently. These relationships and the importance of considering both the distribution of photoelectron energies produced by ATI and the coupling of electrons to the fundamental field can perhaps best be illustrated by Fig. 3, which shows the influence of the photoelectron order on both relative photoelectron production and the strength of electron-photon field coupling. The former is given by the matrix element T^{KFR} [Eq. (33)] and the latter is qualitatively illustrated by the right-hand side of Eq. (56). The logarithmic decline of the electron production rate with increasing j, combined with the comparatively rapid rise in the strength of the electron-photon field coupling that is observed for $j\lesssim40$, favors the disproportionate contribution of the lowest few ATI orders to the harmonic-generation process. This conclusion will be underscored in the comparisons of QED theory with experiment to follow.

Consequently, cutoff for harmonic spectra depends both upon the laser intensity and the magnitude of the electron momentum (e.g., j). From a mathematical perspective, cutoff arises from the property of the ordinary Bessel function $J_j(x)$ that its value drops sharply when j exceeds the argument x. Given the definition of the generalized Bessel function $\mathcal{J}_j(x,y,\phi)$ in terms of ordinary Bessel functions [Eq. (34)], one concludes that cutoff for each generalized Bessel function occurs when $j\approx|x|+|2y|$. Comparing Eqs. (26), (35), (36), (47), (48), and (51), therefore, the cutoff harmonic of a spectrum for a specific value of j can be expressed as

$$q_{\mathrm{cutoff}}\approx 2z+4\left[z\left(j-z-\frac{E_b}{\omega}\right)\right]^{1/2}. \tag{58}$$

In other words, each ATI photoelectron order has an associated spectrum whose cutoff is given by the above relation. Each ATI order correlates with a specific (discrete) photo-

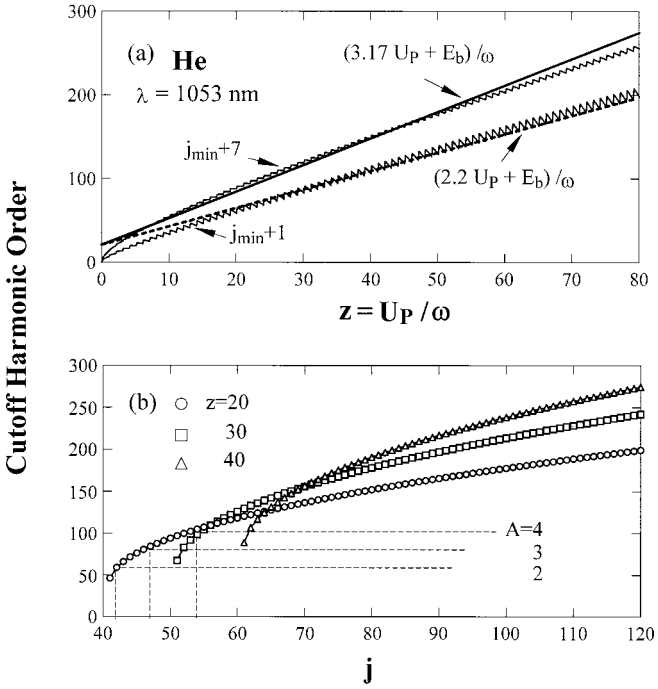


FIG. 4. Variation of the cutoff harmonic order with z $(\equiv U_p/\omega)$ and j. All of the calculations are for He and a laser wavelength of 1053 nm. Reprinted for convenience from Ref. [44], the upper portion of the figure shows the dependence of cutoff on z for two assumed values of j. Including only the lowest photoelectron orders yields a dependence on z that is well-described by the relation $(2.2U_p+E_b)/\hbar\omega$ (shown by the dashed line) but approaches the semiclassical expression $(3.17U_p+E_b)/\hbar\omega$ for larger values of j. Part (b) illustrates the dependence of the cutoff harmonic order on j for various values of z and corroborates the trends of (a). Assuming $z=20$ $(j_{\min}=41)$, the dashed horizontal lines represent the relation $(AU_p+E_b)/\hbar\omega$ for values of A between 2 and 4.

electron energy and the overall HG spectrum is determined by the degree of participation by each order (to be discussed later). This contrasts with the semiclassical two-step model that requires the electron to enter the continuum with zero energy.

It should be mentioned at this juncture that the physical interpretation of cutoff in the QED picture also differs from that in the semiclassical view. From the QED perspective, cutoff is the result of a sharp decline in the Volkov- to Volkov-state transition probability, whereas, for the latter, cutoff represents an absolute limit imposed by energy conservation. In an effort to illustrate the relationship between the semiclassical and QED predictions regarding cutoff, Fig. 4 shows the variation of the cutoff harmonic order with z and j, calculated for He assuming a laser wavelength of 1053 nm, and compares the results with semiclassical predictions. Part (a) of the figure, reprinted for convenience from Ref. [44], displays the dependence of cutoff on z for several values of j. When only the lowest photoelectron orders are considered, cutoff is well-represented by the relation $(2.2U_p+E_b)/\hbar\omega$ (shown by the dashed line, $E_b=24.6\,\mathrm{eV}$) but approaches the semiclassical expression $(3.17U_p+E_b)/\hbar\omega$ for larger values of j $(\geqslant j_{\min}+7,\ |\vec{P}|^2/2m\gtrsim8\,\mathrm{eV})$. The lower portion of Fig.

4 illustrates the dependence of the cutoff harmonic order on j for a fixed value of z. Results are shown for several values of z (again, for He and assuming $\lambda = 1.053\,\mu$m) and, for convenience, the dashed horizontal lines indicate the cutoff harmonic order given by the relation $AU_p + E_b$, where $z = U_p/\omega = 20$. Notice that if only the lowest two to three photoelectron orders contribute to HG, cutoff will be given approximately by $2U_p + E_b$, whereas, if higher orders participate, the expression for cutoff for $z = 20$ will approach $3U_p + E_b$. Similar conclusions apply for other values of the ponderomotive potential.

Considering the data of Fig. 2, then, the critical issue becomes determining the j values to be incorporated into the QED calculations. At first glance, it would seem to be necessary to sum Eq. (51) over all photoelectron orders (as in Fig. 2) to determine T_q. However, including all values of j in the calculations does not, for the following reasons, often yield a physically meaningful result. As was noted in Sec. II C, photoelectrons that do not eventually return to the parent atom carry away momentum and energy that prevent them from participating in the harmonic-generation process. Consequently, the duration of the laser pulse and the energy with which an electron is born in the continuum are crucial factors in determining the photoelectron orders that interact most strongly with the optical field and contribute appreciably to HG. Until recently, virtually all HG experiments involved laser pulses typically $\geqslant$150 fs in duration, which is sufficiently long to permit the higher-energy ATI photoelectrons to escape from the parent atom during the pulse risetime. A photoelectron produced in the continuum with an energy of $\sim$9 eV, for example, which would correspond to the 6th photoelectron order when photoionizing Ne at a wavelength of 0.8 μm and an intensity of $1.3 \times 10^{15}\,\mathrm{W\,cm^{-2}}$, travels $\sim$1500 Å in 75 fs. It is desirable, therefore, that the optical field intensity reach its peak value before the electron escapes from the parent atom. ATI experimental results suggest that these conclusions are valid. Mohideen *et al.* [50], for example, detect considerably fewer cold ($<$ 5 eV) photoelectrons in He ATI experiments ($7 \times 10^{15}\,\mathrm{W\,cm^{-2}}$, 180-fs pulses) than expected on the basis of KFR or tunneling theory. Photoelectron energy spectra acquired for high optical fields in other experiments [49,62–67] have consistently shown the comparative suppression of the lowest ATI orders relative to their higher-energy counterparts. The number of the lowest-order photoelectrons grows with increasing optical field intensity much more slowly than do those for the higher orders and this effect correlates well with the emergence of harmonic generation for optical intensities exceeding $\sim 10^{12}\,\mathrm{W\,cm^{-2}}$. In summary, physical considerations, in conjunction with momentum conservation, and experimental observations suggest that, for most of the experiments reported to date, the lowest photoelectron orders contribute disproportionately to harmonic generation.

Returning to Eq. (51), summing over all j is inappropriate when simulating, for example, ''long pulse'' ($\tau \geqslant 100-150$ fs) experiments and, as we shall see, doing so yields results that are at variance with experimental spectra and polarization data. The approach adopted here is to begin with $j_{\min}$ (the minimum number of absorbed photons necessary to photoionize the atom or molecule at a specific intensity) and to incorporate successively higher values of j into the calculation of the transition matrix T_q while comparing the results with experiment and semiclassical theory. In the long pulse case, the calculations demonstrate that including only the lowest two or three photoelectron orders accounts for experimental data.

Decreasing the laser pulse width further, however, is an entirely different matter. Recent experiments [16–20] in which harmonics have been generated for pulse widths $<$30 fs are intriguing in that they have succeeded in effectively freezing the motion of much higher-energy ATI photoelectrons during the pulse risetime, resulting in harmonic spectra comprising contributions from a greater breadth of photoelectron orders. The 9-eV electron mentioned earlier, for example, travels only $\sim$150 Å in 7.5 fs. In this situation, summing over a larger range in j yields spectra that are in agreement with semiclassical predictions and the limited spectral data available to date.

Since the harmonic spectra will quickly converge beyond a certain value of j ($\equiv j_{\mathrm{con}}$) that rises with increasing intensity, maximizing production of the highest-order harmonics compels us to increase the laser intensity while simultaneously decreasing the pulse width. That is, as the peak intensity of the pulse rises, the number of photoelectron orders required to reach convergence increases, which, in turn, implies a shorter laser pulse width to effectively freeze the motion of these photoelectron orders during the laser pulse. Said another way, for a given laser intensity, reducing the laser pulse width to the point at which the photoelectron orders from $j_{\min}$ to j_{con} are effectively utilized produces the maximum benefit in terms of short-wavelength production. Further reductions in laser pulse width for a fixed intensity yield marginal improvements in harmonic production.

With these considerations in mind, we now turn to a comparison of the present theory with semiclassical predictions and experimental data in the literature.

B. Harmonic spectra: Impact of U_p

Extensive comparisons of QED theory with both semiclassical theory and experimental data have been carried out and representative results are presented in this section. We limit our comparison with existing theory to semiclassical treatments in which the time-dependent Schrödinger equation (TDSE) is solved because other theories generally possess parameters that are not related to physical observables. Figure 5, reprinted from Ref. [44] for convenience, compares QED theoretical spectra [calculated from Eq. (55)] with harmonic spectra calculated by Krause *et al.* [34] by treating the optical field classically and numerically integrating the TDSE. Results are shown for two fundamental optical field wavelengths (λ), 527 and 1053 nm, and at least three sets of j values. The largest set of j values given in both portions of the figure indicate the minimum number of photoelectron orders required to obtain convergence of Eq. (51). Therefore, the spectra associated with these sets of j are essentially identical to the $j \to \infty$ limit. For all of the calculations, the laser

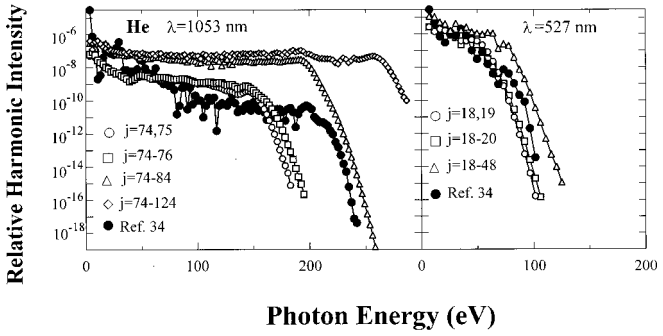


FIG. 5. Comparison of harmonic spectra reported in Ref. [34] for He with those calculated from Eq. (55) for several sets of j values. For a fundamental optical field wavelength (λ) of 527 nm, $j_{\min}=18$ and results for $j=18,19$ ($\bigcirc$), $j=18-20$ ($\square$), and $j=18-48$ ($\triangle$) are shown. For $\lambda=1053$ nm, $j_{\min}=74$ and spectra for $j=74,75$ ($\bigcirc$), $j=74-76$ ($\square$), $j=74-84$ ($\triangle$), and $j=74-124$ ($\diamond$) are presented. The laser intensity for all calculations is $6\times10^{14}\,\mathrm{W\,cm^{-2}}$.

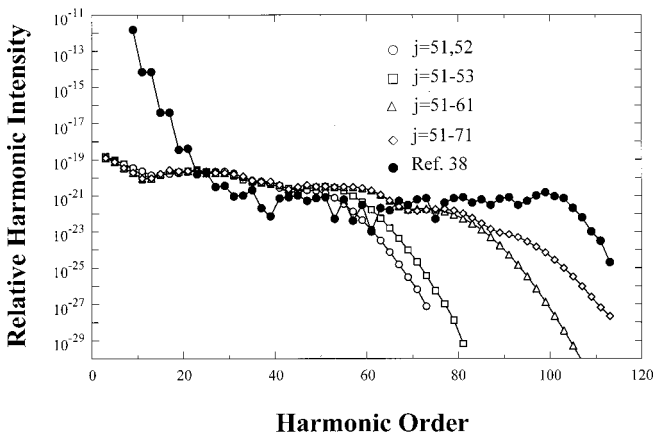


FIG. 6. Harmonic spectrum, calculated by Lewenstein *et al.* (Ref. [38], Fig. 6) from semiclassical theory, compared with QED theory. For the latter, $j_{\min}=51$ and results for four sets of j values are shown: $j=51,52$ ($\bigcirc$); $j=51-53$ ($\square$), $j=51-61$ ($\triangle$); and $j=51-71$ ($\diamond$). Convergence of Eq. (51) occurs for $\Delta j=20$ (i.e., $\boldsymbol{j}=51-70$) for $U_p=20$ eV and $E_b=30$ eV. All of the spectra were normalized arbitrarily at $q=21$.

intensity was taken to be $6\times10^{14}\,\mathrm{W\,cm^{-2}}$. Note that, for the shorter fundamental field wavelength, the ponderomotive potential $U_p(\sim15.5\,\mathrm{eV})$ is smaller than the ionization potential of He ($E_b\approx24.6\,\mathrm{eV}$), QED theory matches the spectrum of Ref. [34] quite well, and the spectra are relatively insensitive to the number of photoelectron orders included in the calculations. For $\lambda=1053\,\mathrm{nm}$, however, $U_p\simeq62\,\mathrm{eV}$ and the two theoretical approaches yield considerably different results. Including all ATI photoelectron orders results in a spectrum whose cutoff lies at approximately the 219th harmonic ($\hbar\omega\approx260\,\mathrm{eV}$) or $\sim$18% higher than that for the spectrum of Ref. [34] ($q\simeq186$, $\hbar\omega\sim220\,\mathrm{eV}\simeq3.15U_p+E_b$). If one considers only the lowest two or three photoelectron orders, the QED spectrum cutoff is $\sim148\,\mathrm{eV}\approx2U_p+E_b$, or $\sim$48% lower than that associated with the semiclassical calculation. Matching the cutoff of the spectrum of Ref. [34] requires almost ten photoelectron orders ($j_{\min}=74\rightarrow84$) but the QED spectra, regardless of the range in j, do not decline in intensity with increasing q as rapidly as does the spectrum calculated by Krause *et al.* [34]. These results illustrate a recurring theme when comparing QED theory with semiclassical predictions and experiment. With few exceptions, including all ATI orders in the QED calculations matches neither experiment nor semiclassical theory. Including only two or three photoelectron orders, on the other hand, reproduces virtually all of the experimental data available in the literature and is consistent with the physical considerations discussed in Sec. III A.

Calculations reported by Lewenstein *et al.* [38] for an atom whose ground-state wave function is given by $\Phi_i(r)=(\alpha/\pi)^{3/4}\exp(-\alpha r^2/2)$ (where $\alpha=30$) are compared in Fig. 6 with the theory presented here. All of the calculations assume that $E_b=30\,\mathrm{eV}$ and $U_p=20\,\mathrm{eV}$ and the spectra have been arbitrarily normalized at $q=21$. The relative intensity of the semiclassical spectrum [38] falls in a perturbative manner by eight orders of magnitude as q increases from 9 to 23, and remains essentially constant over more than 60 harmonic orders. In contrast, the QED calculations predict a more gradual decline in harmonic intensity and cutoff at $q\approx53$ and 77 for the $j=51-53$ ($\square$) and $j=51-61$ ($\triangle$) spectra, respectively. For the parameters of Fig. 6 mentioned above, convergence of Eq. (51) occurs when the first 20 orders are included in the calculations and the resulting spectrum is illustrated by the open diamonds ($\diamond$). The harmonic response predicted by QED, even in the $j\rightarrow\infty$ limit, is considerably weaker than the semiclassical spectrum above $q\approx80$ and the two spectra differ in relative intensity by almost three orders of magnitude at the harmonic cutoff for the semiclassical spectrum ($q\approx101$). More comparisons with semiclassical theory will be presented later in connection with the discussion of theoretical predictions and experimental data for short ($<$50 fs) laser pulses.

Comparison of QED theory with experimental harmonic spectra must necessarily be somewhat qualitative at this stage because, as noted earlier, propagation of the single atom response through the gaseous medium has not yet been incorporated into the calculations and, hence, phase-matching effects are not taken into account. Nevertheless, the comparisons presented here are adequate for demonstrating the predictive capability of QED theory. In Fig. 3 of Ref. [44], the harmonic spectrum acquired by Macklin *et al.* [11] was compared with QED predictions and it was shown that, for intensities within $\pm25\%$ of the published value ($1.3\times10^{15}\,\mathrm{W\,cm^{-2}}$) [11], the QED spectra are consistent with experiment if only the lowest two or three photoelectron orders are included in Eq. (51). A similar result is obtained if we now consider the data reported by L'Huillier and Balcou [12]. The results of Ref. [12] were obtained for a fundamental wavelength of 1.053 μm and a peak intensity (''best focus'' value) [12] of $1.5\times10^{15}\,\mathrm{W\,cm^{-2}}$, and Fig. 7 shows their data for Ne. The theoretical predictions favor a lower laser intensity to describe the experimental spectrum. Specifically, calculations for $I=(0.7-1.0)\times10^{15}\,\mathrm{W\,cm^{-2}}$ and the lowest two photoelectron orders ($j=80,81$ for $I=7\times10^{14}\,\mathrm{W\,cm^{-2}}$) agree well with the general characteristics

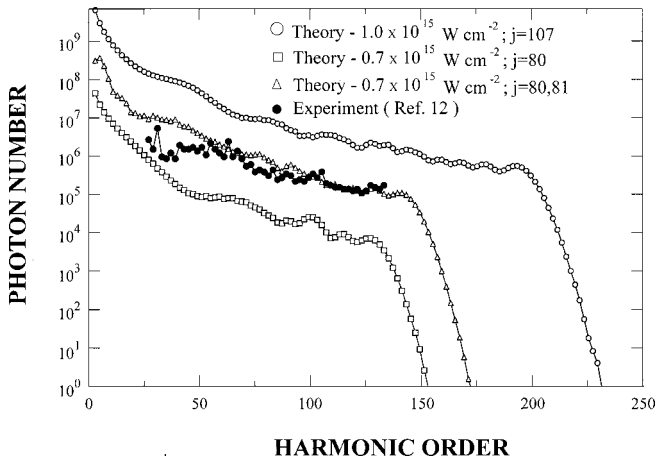


FIG. 7. Comparison of the harmonic spectrum of 1.053 μm in Ne reported in Ref. [12] with QED calculations. The fundamental optical field intensity in Ref. [12] was stated to be $1.5\times10^{15}\,\mathrm{W\,cm^{-2}}$. Three theoretical spectra are shown which correspond to two values of intensity and different sets of j: $I=1\times10^{15}\,\mathrm{W\,cm^{-2}}$, $j=107$ (○); $I=7\times10^{14}\,\mathrm{W\,cm^{-2}}$, $j=80$ (□); $I=7\times10^{14}\,\mathrm{W\,cm^{-2}}$, $j=80,81$ (△).

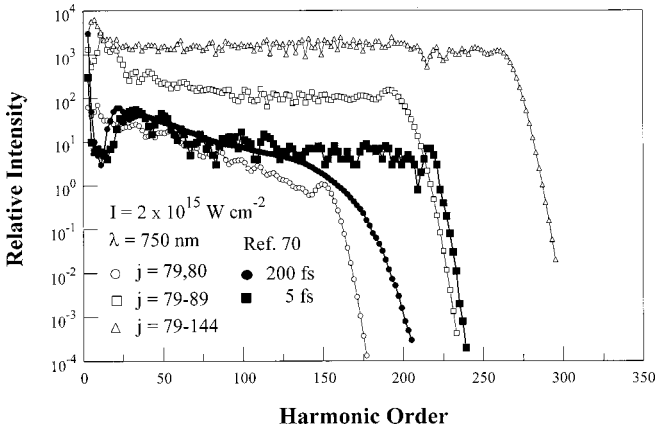


FIG. 8. Harmonic spectra for He calculated from semiclassical theory (Ref. [38]) by Kan *et al.* (Ref. [70]), compared with the results from QED theory. The spectra from Ref. [70] assume laser pulse widths of 5 fs (■) and 200 fs (●), whereas the spectra from Eq. (55) are independent of pulse width but calculations are shown for several sets of j values. For all spectra shown, the laser intensity and wavelength are assumed to be $2\times10^{15}\,\mathrm{W\,cm^{-2}}$ and 750 nm, respectively. Under these conditions, Eq. (51) converges when 66 photoelectron orders ($j=79-144$) are included. For illustrative purposes, the structure in the 5-fs spectrum of Ref. [70], especially that near cutoff, is not reproduced here.

of L'Huillier and Balcou's results, although cutoff was apparently not observed in the experiments. If cutoff for the experimental data is assumed to occur immediately beyond the highest-order harmonic reported, then including all photoelectron orders cannot be reconciled with the data of Ref. [12] for intensities within a factor of 3 of the stated value $[1.5\times10^{15}\,\mathrm{W\,cm^{-2}}$ (cf. Fig. 5)]. Also, cutoff for the $7\times10^{14}\,\mathrm{W\,cm^{-2}}$, $j=80$ (□) and $1\times10^{15}\,\mathrm{W\,cm^{-2}}$, $j=107$ (○) spectra of Fig. 7 occurs for $q\simeq129$ and 195, respectively, which are consistent with the expression for q_{cutoff} given earlier. When only the lowest photoelectron orders ($j\approx j_{\mathrm{min}}$) are included in the calculation of harmonic spectra, then, as discussed previously in connection with Figs. 4 and 5, Eq. (57) yields $q_{\mathrm{cutoff}}\approx2z$, which is consistent with the theoretical and experimental data of Fig. 7 as well as the experimental observations reported in Refs. [13] and [68], and the comparison of the data of Macklin *et al.* [11] with QED theory in Fig. 3 of Ref. [44]. In summary, not only does the theory presented here point toward the utilization of only the $j_{\mathrm{min}}\rightarrow j_{\mathrm{min}}+2$ photoelectrons in the HG process for laser pulse widths $>100-150$ fs, but agreement with experimental spectra generally requires intensities that are 10–50 % lower than the stated peak value. Also, although propagation and phase-matching effects are capable of altering the harmonic spectral profile, the comparisons presented here and in Ref. [44] illustrate that the QED framework yields spectra that are in qualitative agreement with experimental results and semiclassical predictions.

Now we turn our attention to the harmonic spectra that have recently been calculated and recorded for laser pulse widths as small as ~5 fs. Advances in the technology of mode-locked Ti:Al_2O_3 oscillator/amplifier systems provide for the generation of pulses having durations of only a few optical cycles in the near-infrared while maintaining focal intensities in the range of $(4-6)\times10^{15}\,\mathrm{W\,cm^{-2}}$ (Refs. [16–20]). Decreasing the laser pulse width has been observed to yield higher-energy harmonics [16,69]. Chang *et al.* [16], for example, report harmonics as high as $q=155$ and 221 when Ne and He, respectively, are irradiated with ~26-fs pulses, whereas the generation of only the 119th and 239th harmonics are expected (for the same fundamental intensity) when the laser pulse width is 100 fs.

However, experimental observations of harmonic spectra generated by pulse widths below 50 fs have, to date, been few and the difficulties associated with acquiring spectra over a broad region in the XUV and soft-x-ray regions that are corrected for the response of the detection system, including filters and gratings, are well known. Accordingly, this section will focus on both semiclassical predictions of HG spectra produced by sub-50-fs pulses and recent experimental results. A comparison of spectra calculated (in the absence of propagation effects) by Kan and co-workers [70] with their QED counterparts is presented in Fig. 8. Calculated for He from the theory of Lewenstein *et al.* [38] for laser pulse widths of 5 and 200 fs and a fundamental wavelength and intensity of 750 nm and $2\times10^{15}\,\mathrm{W\,cm^{-2}}$, respectively, the semiclassical spectra exhibit a pronounced extension towards shorter wavelengths as the laser pulse width is reduced, an effect which has been partially attributed to a reduction in the fractional ionization of the rare gas before peak laser intensity is reached. The QED spectra shown in Fig. 8 are, on the other hand, not explicitly dependent on laser pulse width but results are again shown for several sets of j values while assuming the same fundamental laser wavelength and intensity as those for the semiclassical calculations. It is clear that the QED spectrum for $j=79,80$ is remarkably similar to the long pulse (200 fs) semiclassical spectrum. Notice that the slopes of the QED and semiclassical spectra in the plateau are almost identical and, although

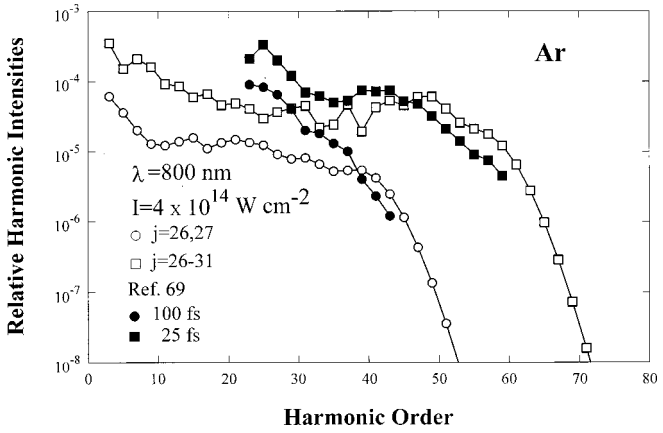


FIG. 9. Experimental harmonic spectra reported in Ref. [69] for laser pulse widths of 100 fs (●) and 25 fs (■), compared with the present theory for two sets of j values and $j_{\min}=26$. The spectra shown are for Ar and a laser wavelength and intensity of 800 nm and $4\times10^{14}\,\mathrm{W\,cm^{-2}}$, respectively. For these parameters, convergence occurs when $\Delta j\cong26$ (i.e., $j=26-51$).

the $j=79,80$ QED spectrum declines more sharply beyond $q\approx150$ than does the semiclassical spectrum, both cut off at roughly the same photon energy. Similarly, the 5-fs spectrum of Ref. [70] is quite close to its $j=79-89$ QED counterpart. Again, the plateau regions have approximately the same slope and the shapes of both spectra in the vicinity of cutoff closely resemble one another. This is entirely consistent with the picture discussed earlier in which reducing the laser pulse width has the effect of momentarily trapping photoelectrons beyond the lowest few orders. Because of their stronger coupling with the photon field, these photoelectrons significantly extend the harmonic response to shorter wavelengths. If this picture is valid, we note, however, that even the spectrum associated with 5-fs pulses does not represent the limit achievable for this intensity because the open triangles in Fig. 8 illustrate the spectrum associated with $j=79-144$, the range of j values required for Eq. (51) to converge. Therefore, further improvements in short-wavelength generation can be expected for this intensity as the pulse width is reduced below 5 fs.

The same trends are observed when comparing the QED theory with experimental results. Consider, for example, Fig. 9, which presents a comparison of two of the spectra reported by Christov *et al.* [69] with those predicted by the present theory. Although propagation effects are not included in the QED spectra, it is nonetheless clear that the calculations for the lowest two photoelectron orders ($j=26,27$) yield a spectrum that is similar to that for the 100-fs data of Ref. [69]. This is particularly true in the cutoff region although the experimental data indicate a lower value for the cutoff harmonic. Furthermore, the 25-fs experimental spectrum reflects well the QED result for $j=26-31$. That is, reducing the temporal width of the laser pulse by a factor of 4 while holding the intensity at a constant value increases, by roughly a factor of 3, the number of photoelectron orders contributing significantly to harmonic production. This is consistent with the conclusions reached on the basis of Fig. 8 and comparisons of QED theory with the harmonic spectra calculated by Gaarde *et al.* [71].

Note, too, that Figs. 8 and 9 are in accord with the experimental spectra of Ref. [69] in another important aspect—namely, QED theory predicts that, as the duration of the laser pulse is decreased while maintaining a constant optical field intensity, the addition of higher photoelectron orders affects primarily the cutoff region. As is evident in Fig. 1 of Ref. [44] and Figs. 2, 5, 6, 8, and 9 shown earlier, spectral convergence occurs quickly in the plateau as one adds photoelectron orders to the calculations. Including still higher orders contributes almost exclusively to the harmonic response in the cutoff region, which is the last portion of the spectrum to show convergence of Eq. (51). Therefore, we would expect to see experimentally what has been observed—i.e., decreasing the laser pulse width extends the short-wavelength limit of the spectrum while leaving the plateau relatively unaffected.

In summary, from Figs. 8 and 9 and similar calculations carried out to date, it is clear that the QED view of Volkov-to Volkov-state transitions as the source of HG radiation is consistent with semiclassical calculations, experimental spectra, and limitations on the number of j values included in the calculations that are determined by physical considerations. Our proposal, then, is that the same mechanism for generating high-order harmonics is at work when the laser pulse width is <30 fs as in the ''long pulse'' (>100 fs) regime. Reducing the duration of the fundamental optical field, however, effectively traps or ''freezes'' photoelectrons of increasingly higher energies which, by virtue of the dependence of the optical field-electron coupling on electron energy, results in an extension of the harmonic response to shorter wavelengths. Viewed from another perspective, the time-independent theory presented here is linked with the temporal domain through the energies of the photoelectrons.

C. Polarization data

Following the initial observations of harmonic spectra, it was recognized that several semiclassical models were capable of predicting the general characteristics (i.e., plateau and cutoff) of experimental spectra. Combined with the potential impact of propagation effects on the spectral profile, this situation prompted several groups to suggest that the characteristics of harmonics generated in response to an elliptically polarized fundamental optical field would present a more demanding test of a proposed model. Burnett, Kan, and Corkum [22] noted that ''To a first approximation, propagation and phase matching in the atomic medium are not affected by small variations in [the] ellipticity of the fundamental, while the individual atom nonlinear response can vary greatly. Ellipticity can thus be regarded as a tool, allowing one to isolate the single-atom response to the fundamental field.'' In a similar vein, Weihe *et al.* [23] observe that ellipticity '' . . . experiments are a sensitive probe of the evolution of the radiating dipole at the moment high harmonics are generated'' and Becker *et al.* [43] succinctly summarize the critical role of ellipticity data in evaluating competing theories in stating that '' . . . these effects [the degree of ellipticity and the orientation of the axis of the polarization

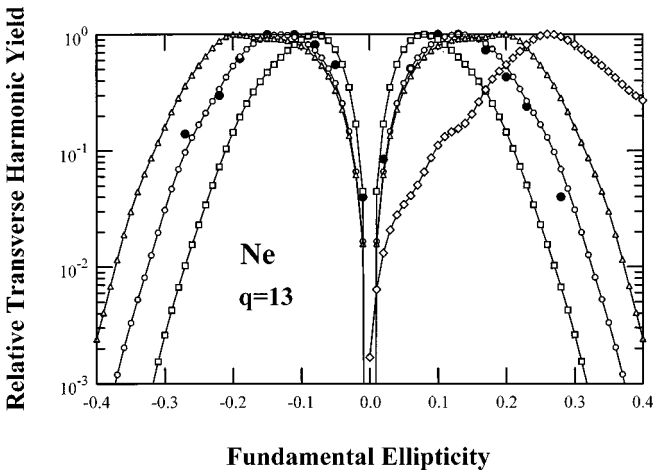


FIG. 10. Comparison of the relative yield of the transverse component of the 13th harmonic of 754 nm in Ne with the experimental data of Ref. [22] (●). The fundamental field intensity is assumed to be $10^{15}\,\mathrm{W\,cm^{-2}}$, and results for four sets of photoelectron orders are shown: $j=46$ (□); $j=46,47$(○); $j=46$–48 (△); and $j=46$–71 (◇).

ellipse] . . . might provide a more stringent test of any theory than just the harmonic intensities.''

For these reasons, a number of experimental studies of the effect of fundamental field ellipticity on harmonic generation have been reported [21–29] and the measurements fall into two broad categories. The first is concerned with the variation of the relative harmonic yield with the degree of fundamental field ellipticity [21,22,24,26–28], whereas the second focuses on the characteristics of the polarization ellipse associated with a given harmonic [23,25,29]. All of the experiments reported to date involved laser pulse widths $\geqslant$140 fs. Several theoretical approaches to describing harmonic generation in the presence of an elliptically polarized fundamental optical field have appeared in the literature [28,43,72], all of which treat the interaction of the electron with the laser field in a classical manner. Antoine *et al.* [28,29] also include propagation effects in the calculations and compare several results with experiment. In a number of instances, phase matching is observed to have a significant impact on the theoretical predictions. However, the calculations of harmonic yield were found to be relatively insensitive to propagation and, in particular, for the lower harmonics.

Recently, we reported [73] a comparison of QED theory with a sampling of the available polarization data and more results are presented here which demonstrate the remarkable degree to which Eq. (51) reproduces experimental data. Let us first consider experiments in which the relative yield of a specific harmonic has been measured. Figure 10 gives a comparison between theory and the experimental data of Burnett *et al.* [22], who measured the transverse component of the relative yield of the 13th harmonic of 754 nm in Ne. Measurements [22] of the components of the harmonic radiation that are parallel or transverse to the major axis of the polarization ellipse for the fundamental show (cf. Fig. 10) that the transverse yield essentially vanishes for zero ellipticity and reaches its peak value for $0.1<|\mu|<0.2$. Although not shown in Fig. 10, the *parallel* yield is roughly constant for $|\mu|<0.1$. The theoretical results presented in Fig. 10, most of which were published previously [73], were calculated by evaluating the expressions

$$Y_y^q \propto \frac{1}{2\pi}\int_0^{2\pi} |T_q(\hat{\varepsilon}'^* = \hat{\varepsilon}_y)|^2 d(\Delta\Theta) \tag{59}$$

and

$$Y_x^q \propto \frac{1}{2\pi}\int_0^{2\pi} |T_q(\hat{\varepsilon}'^* = \hat{\varepsilon}_x)|^2 d(\Delta\Theta) \tag{60}$$

for the transverse and parallel components, respectively, of the polarization ellipse for the qth harmonic. It is evident that including only the lowest two photoelectron orders ($j=46$ and 47, where $j_{\min}=46$ for $I=10^{15}\,\mathrm{W\,cm^{-2}}$ and $\lambda=754\,\mathrm{nm}$) reproduces the data over more than an order of magnitude in relative yield (intensity) but including higher-energy photoelectrons results in a rapid departure from experimental observations. Specifically, if we include the lowest 20 photoelectron orders ($j=46$–66), which are necessary to reach convergence of Eq. (51), it is clear that the maximum transverse response is predicted to lie at $\mu\sim0.26$ and is at odds with the data. As was noted previously, all of the polarization data presently available were acquired for fundamental optical pulse widths well above 100 fs. Thus, we have yet to see the impact of a shorter laser pulse on data such as those of Fig. 10. The agreement between the experimental results and the $j=46,47$ calculations are again attributed to the escape of the higher-energy photoelectrons ($j>47$) during the pulse risetime. This interpretation and the $j=46$–66 curve of Fig. 10 predict that measurements similar to those of Ref. [22] carried out with shorter laser pulse widths ($<$50 fs) will show the maximum transverse yield lying at larger values of μ.

In Ref. [73], measurements [21,28] of the total harmonic yield as a function of ellipticity for various harmonics were compared with QED and semiclassical theory. A similar comparison is given in Fig. 11, which shows calculated harmonic yield as a function of μ for $\lambda=754\,\mathrm{nm}$ and $I=10^{15}\,\mathrm{W\,cm^{-2}}$ for several harmonic orders. The experimental data of Refs. [22] and [26] are also included. For $q=15$ and 17, the relative yields of Refs. [22] and [26] are generally well within a factor of 2 (and a factor of 3 for all of the harmonics) of the theoretical predictions. These data support the same conclusion as that reached in connection with Fig. 10—the experimental observations are well-explained if one incorporates only the lowest two or three photoelectron orders into the calculations, whereas higher orders result in extending the harmonic response to larger values of ellipticity. Also, the sensitivity of the yield (transverse and total) calculations to the number of photoelectron orders affords a valuable diagnostic of the HG process.

Although Eqs. (51), (55), (59), and (60) reproduce the variation of the parallel and perpendicular components of the harmonic yield with ellipticity, the theory predicts *relative* yields for the parallel component that are approximately two orders of magnitude larger than those for the perpendicular

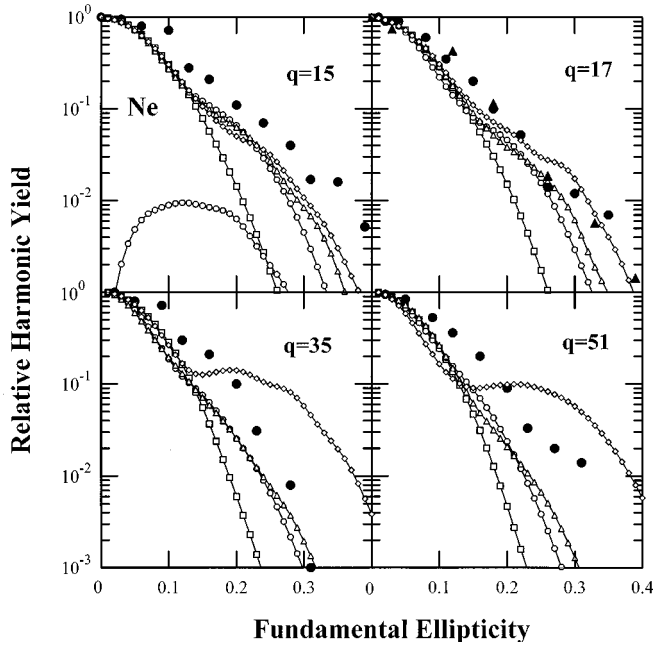


FIG. 11. Dependence on the fundamental ellipticity of the relative yield for the $q=15$, 17, 35, and 51 harmonics of 754 nm in Ne for an intensity of $10^{15}\,\mathrm{W\,cm^{-2}}$. The ATI orders displayed are: (□) $j=46$; (○) $j=46,47$; (△) $j=46-48$; (◇) $j=46-49$. The experimental data of Refs. [22] (●) and [26] (▲) are also shown. The $q=15$ panel includes the calculated perpendicular component of the harmonic yield for $j=46,47$ from Fig. 10.

component. The upper left ($q=15$) panel of Fig. 11 includes the perpendicular component calculations from Fig. 10 for $j=46,47$ as well as the total harmonic yield. Since the experiments of Ref. [22] show the relative magnitudes of the perpendicular and parallel portions of the harmonic yield for $q=15$ to be comparable for ellipticities between ~0.05 and 0.25, then either the theory incorrectly predicts a smaller perpendicular yield *or*, for several harmonics having photon energies ($q\omega$) near the atomic ionization potential, the parallel polarization is preferentially attenuated by the surrounding gas through absorption and nonlinear effects, thereby exposing the underlying perpendicular contribution. It appears that the latter is at least partially the case. Burnett and coworkers [22] note that the comparable magnitudes of the components of the 15th harmonic polarized parallel and transverse to the driving optical field is a situation not observed in general: ''In contrast to the [15th] harmonic, the [17th] harmonic was largely polarized parallel to the major polarization axis of the fundamental for [ellipticities] in the range 0–0.1 . . . a weak component . . . was observed in the transverse polarization . . . '' [22]. We conclude that the total yield for most of the harmonics conceals the interesting behavior of the perpendicular component because of the relative strength of the parallel component. Only for photon energies near the ionization potential, where a variety of single photon absorption and multiwave mixing processes [24,27] occur, is the dip in the total harmonic yield noticeable in the vicinity of zero ellipticity. It should be emphasized, however, that these processes are unrelated to the primary mechanism by which harmonics are generated. In summary, it appears

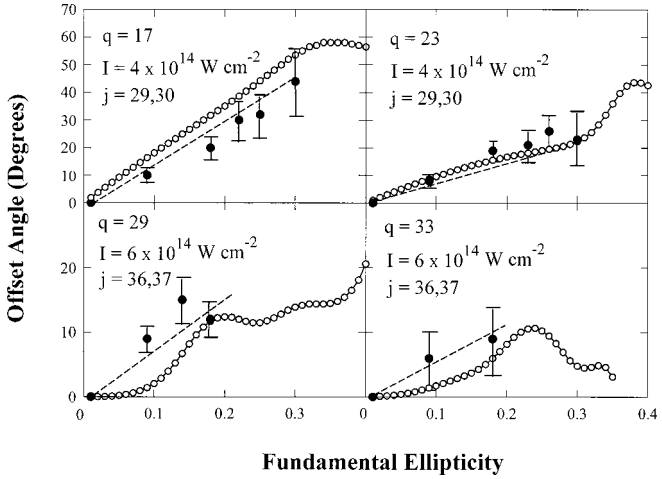


FIG. 12. Calculations of the offset (rotation) angle for the 17th, 23rd, 29th, and 33rd harmonics of 790 nm in Ne at fundamental field intensities of $(4-6)\times10^{14}\,\mathrm{W\,cm^{-2}}$. Results for two sets of photoelectron orders ($j=29,30$ for $q=17$ and 23, $I=4\times10^{14}\,\mathrm{W\,cm^{-2}}$; $j=36,37$ for $q=29$ and 33, $I=6\times10^{14}\,\mathrm{W\,cm^{-2}}$) are shown and the calculations are compared with semiclassical theory (dashed line) and experimental data (●) from Ref. [29]. Note the change in the scale of the ordinate between the upper and lower portions of the figure.

that the transverse component of the harmonic yield is normally obscured by its stronger counterpart and its dependence on ellipticity is obvious from the total harmonic yield only for harmonics having photon energies comparable to the ionization potential.

Next, we turn to an examination of the characteristics of the polarization ellipse for a specific harmonic, whose polarization vector can be expressed as

$$\hat{\varepsilon}'^{*}=\frac{T_q(\hat{\varepsilon}'^{*}=\hat{\varepsilon}_x)\hat{\varepsilon}_x+T_q(\hat{\varepsilon}'^{*}=\hat{\varepsilon}_y)\hat{\varepsilon}_y}{\sqrt{T_q^2(\hat{\varepsilon}'^{*}=\hat{\varepsilon}_x)+T_q^2(\hat{\varepsilon}'^{*}=\hat{\varepsilon}_y)}} \tag{61}$$

for the qth harmonic. For a given ellipticity of the fundamental optical field, the harmonic radiation will, in general, also be elliptically polarized *and* its polarization ellipse may be rotated relative to the ellipse associated with the fundamental. The first experiments in this regard were published by Weihe and co-workers [23,25], who measured the ellipticity of the harmonic radiation as well as the offset (or rotation) angle—the angle between the major axis of the harmonic's polarization ellipse and that for the fundamental. Measurements were made in both Ar and N_2 as a function of the ellipticity of the fundamental field and, in Ref. [73], offset angle data for the 7th–13th harmonics of 785 nm in Ar were compared with QED theory. Extensive measurements of the rotation angle and ellipticity of harmonics of 790 nm in Xe, Ar, and Ne were reported recently by Antoine *et al.* [29]. Taken over a range of harmonics in both the plateau and cutoff regions of the spectrum, these data provide a convenient benchmark against which the QED formalism, and the proposed importance of Volkov→Volkov-state transitions, in particular, may be tested. One comparison between the harmonic ellipticity data of Ref. [29] and QED theory (for

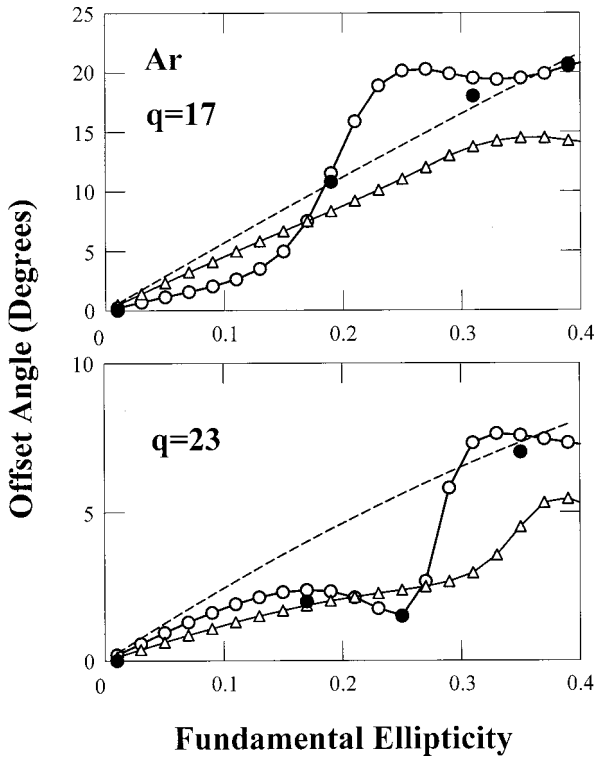


FIG. 13. Data similar to those of Fig. 12 for $q=17$ and 23 in Ar. QED calculations are given for two values of intensity (○, $2\times10^{14}\,\mathrm{W\,cm^{-2}}$; △, $1.5\times10^{14}\,\mathrm{W\,cm^{-2}}$) and including the two lowest photoelectron orders ($1.5\times10^{14}\,\mathrm{W\,cm^{-2}}$: $j=16,17$; $2.0\times10^{14}\,\mathrm{W\,cm^{-2}}$: $j=18,19$). The data and semiclassical theory of Ref. [29] are denoted by the solid circles (●) and the dashed curves, respectively. To compare with the experimental data, the QED results were reduced in magnitude by a factor of 4 and 8 for the $q=17$ and 23 calculations, respectively. The fundamental wavelength is 790 nm.

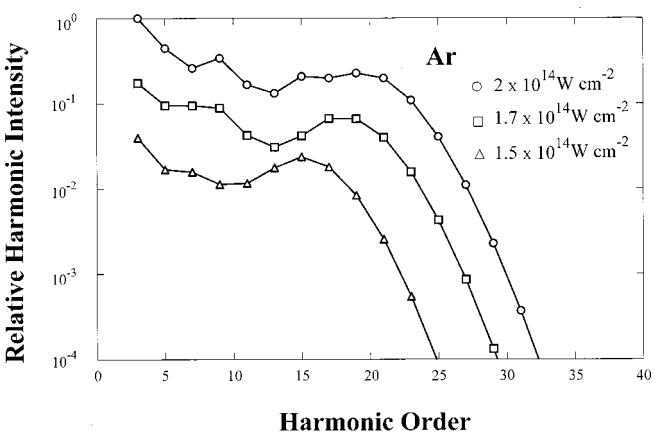


FIG. 14. Harmonic spectra in Ar for a laser wavelength of 790 nm and three values of intensity. All calculations incorporate the lowest two photoelectron orders. The observation of Ref. [29] ($q=17$ in the plateau, $q=23$ in the cutoff region) is met for $I=2\times10^{14}\,\mathrm{W\,cm^{-2}}$.

the 17th and 23rd harmonics of 790 nm in Ne), given by Gao *et al.* [73], suggested that reproducing the experimental data is, in contrast to the rotation angle data, not sensitive to the range of photoelectron orders.

Additional comparisons of QED theory with the results of Ref. [29] are presented in Figs. 12–15. Figure 12 shows the calculated offset angle for the 17th, 23rd, 29th, and 33rd harmonics of 790 nm in Ne and its predicted dependence on fundamental ellipticity, μ. In order to compare theory with the measurements of Antoine *et al.* [29], the calculations for $q=17$ and 23 correspond to a laser intensity of $4\times10^{14}\,\mathrm{W\,cm^{-2}}$ while those for $q=29$ and 33 were carried out for an intensity of $6\times10^{14}\,\mathrm{W\,cm^{-2}}$. Once again, the calculations that best fit the data include only the two lowest photoelectron orders. Because the pulse width of the laser for the experiments of Ref. [29] was ~150 fs, incorporating only one or more than three photoelectron orders into the computations resulted in significant departures from the data. For the lower harmonics (17 and 23), the QED theory agrees quite well with both the semiclassical theory and experimental data of Antoine *et al.* over the range of ellipticity studied in Ref. [29]. Beyond $\mu\sim0.3$, however, QED theory predicts a leveling-off of the offset angle for $q=17$ but a significant rise for $q=23$ that is somewhat akin to that observed for calculations in Ar (to be discussed later). As the harmonic order and laser intensity are increased, however, deviations between the quantum and semiclassical calculations become apparent. The semiclassical theory and data for $q=29$, for example, are in agreement (at least up to $\mu\sim0.18$) but the QED calculations lie outside the error bar for the $\mu\sim0.09$ data point [29]. However, the three data points shown in the $0.1\lesssim\mu\lesssim0.18$ interval imply that the offset angle has reached a local maximum which is the same behavior predicted by the QED calculations. Shifting the QED curve downward in μ by ~0.04 would match the data closely. Above $\mu\sim0.2$, the rate of rise of the semiclassical curve decreases whereas QED theory predicts that the rotation angle remains essentially constant up to approximately $\mu=0.35$. Further experimental measurements in this region would be of benefit in clarifying potential differences between QED and semiclassical predictions. The situation for $q=33$ is similar. Both theories predict small offset angles ($\lesssim10°$) for fundamental ellipticities $\lesssim0.2$ but, in this case, the experimental uncertainty for $\mu\approx0.1$ is apparently larger than that for $q=29$ and the QED curve falls within the error bars for the points reported in Ref. [29]. However, the semiclassical curve appears to continue to rise beyond $\mu\sim0.25$, whereas QED theory predicts that the offset angle quickly reaches a maximum value of ~11° at $\mu=0.23$, declines thereafter, and approaches zero at $\mu\sim0.36$.

Further comparisons of the calculated behavior of the offset angle as a function of μ with the experimental data of Ref. [29] are presented in Fig. 13 for the 17th and 23rd harmonics of 790 nm in Ar, the only two harmonics for which data were reported. All calculations included the lowest two photoelectron orders ($j=j_{\min},j_{\min}+1$) and results for laser intensities of $1.5\times10^{14}\,\mathrm{W\,cm^{-2}}$ and $2\times10^{14}\,\mathrm{W\,cm^{-2}}$ are shown. Antoine *et al.* [29] specified the intensity as $2\times10^{14}\,\mathrm{W\,cm^{-2}}$ and, indeed, the QED theoretical curves reproduce the dependence of the offset angle on the fundamental ellipticity, μ, for both harmonics. The theoretical curve

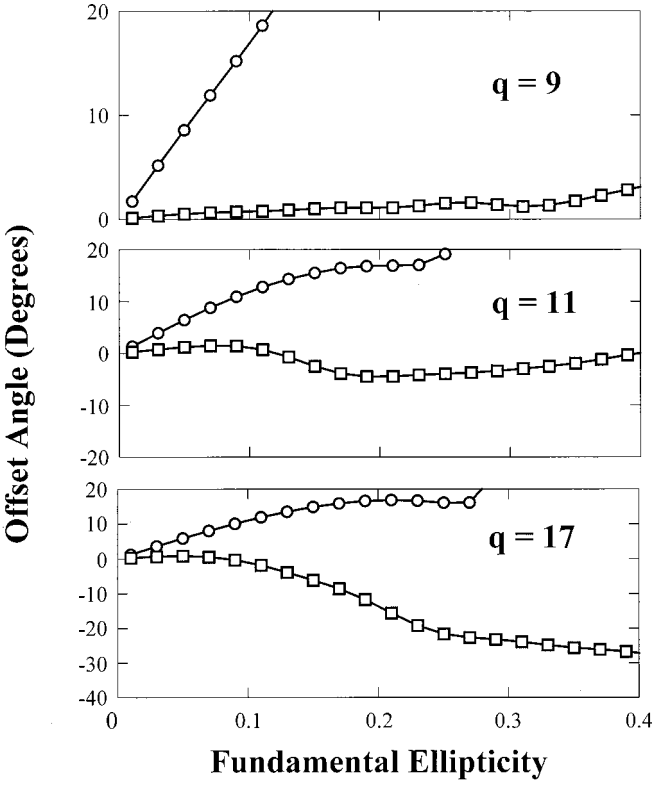


FIG. 15. Dependence of the offset angle on fundamental ellipticity for three harmonics ($q=9$, 11, and 17) of $\lambda=790$ nm in Ar. Results for two values of intensity, $10^{14}\,\mathrm{W\,cm^{-2}}$ (○) and $5\times10^{14}\,\mathrm{W\,cm^{-2}}$ (□), are shown.

for $q=17$ suggests that the offset angle may be saturating for $\mu\gtrsim0.22$. Similarly, the marked rise in offset angle above $\mu\approx0.25$ that is evident in the $q=23$ data is matched by the quantum theory. We note, however, that the *magnitude* of the offset angle for these conditions is not predicted well by the present theory. Although the theory matched the Ne data of Fig. 12 in both magnitude and functional dependence, the QED theoretical curves of Fig. 13 for $q=17$ and 23 have been scaled down by factors of 4 and 8, respectively.

Nevertheless, the theory presented here accurately predicts the dependence of rotation angle on μ and either an increase in the number of photoelectron orders or a variation in the laser intensity results in a rapid deterioration in the quality of the fit to experimental data. The latter is illustrated by the $I=1.5\times10^{14}\,\mathrm{W\,cm^{-2}}$ calculations of Fig. 13, which do not follow the $q=23$ data, for example. Notice that semiclassical theory [29] also predicts a departure from a linear dependence of the offset angle on μ but the deviation from linearity is slight.

In discussing their data, Antoine *et al.* state that ''In the case of argon, at an intensity of $2\times10^{14}\,\mathrm{W\,cm^{-2}}$, the 17th and 23rd harmonics are, respectively, at the end of the plateau and in the cutoff'' [29]. As illustrated in Fig. 14, this condition is met precisely by our calculations of the harmonic spectra. For $I=1.5\times10^{14}\,\mathrm{W\,cm^{-2}}$, both $q=17$ and 23 are in the cutoff region but, at $2\times10^{14}\,\mathrm{W\,cm^{-2}}$, the two harmonics straddle the cutoff harmonic ($q=21$). At still higher intensities, $q=23$ moves into the plateau. We view these results as providing further support for the present theory—namely, that upon specifying the value of intensity on the basis of one experimental observation (spectra), calculations of another observable (polarization offset angle) are also in agreement with experiment.

Another aspect of the results of Fig. 13 that should be mentioned is that, for a fundamental intensity of $2\times10^{14}\,\mathrm{W\,cm^{-2}}$, the offset angle for $q=9$ and 11 (not shown) has the same sign as that for $q=17$ and 23 (and higher harmonics). One may recall that this behavior is quite different from that observed at higher intensities [23,25,73]. Weihe *et al.* [23,25] reported in 1995 and 1996 offset angle data for a fundamental wavelength of 785 nm in Ar showing a reversal in the sign of the offset angle between $q=9$ and 11. In Ref. [73], this change in sign was shown to be predicted by QED theory for laser intensities above $\sim3\times10^{14}\,\mathrm{W\,cm^{-2}}$ but, to our knowledge, has not been reproduced by other theoretical approaches. Figure 15 shows the variation in the offset angle-ellipticity data for $q=9$, 11, and 17 for intensities of $10^{14}\,\mathrm{W\,cm^{-2}}$ and $5\times10^{14}\,\mathrm{W\,cm^{-2}}$. The calculations include only the lowest two photoelectron orders and clearly corroborate the experimental observations that, when $I=10^{14}\,\mathrm{W\,cm^{-2}}$, the offset angles for all three harmonics have the same sign. As the intensity is increased to $5\times10^{14}\,\mathrm{W\,cm^{-2}}$, however, the offset angle takes on negative values for $q=11$ and 17 whereas the $q=9$ data remain positive. In short, the calculations presented here (and Ref. [73]) predict that the offset angles for an intensity of $10^{14}\,\mathrm{W\,cm^{-2}}$ do not change in sign between $q=9$ and 11, whereas QED theory and experiment agree that this is precisely what occurs at higher intensities.

On the basis of calculations carried out to date, of which representative results have been presented here and in Ref. [73], we conclude that QED theory matches the available data remarkably well. Although it must be remembered that propagation effects have not yet been included in the theory, the agreement between QED theory and the harmonic yields, ellipticity, and offset angle data for Ne and Ar are indications that this theory, though differing considerably from the semiclassical view of the HG process, is capable of accurately describing experimental observations.

IV. CONCLUSIONS

A QED picture of the harmonic-generation process that is at once intuitive and yet comprehensive has been described and an analytical expression for the harmonic-generation transition matrix has been derived. Detailed comparisons of the theory with available experimental data show remarkable agreement between the two and the framework presented here reproduces several experimental results that (to our knowledge) have not as yet been described by existing theories. A few differences with experiment remain which are expected to be remedied by incorporating propagation effects and, for harmonics having photon energies comparable to the atomic ionization potential, the Coulombic interaction.

At the core of this theory is the quantum Volkov state and Volkov→Volkov state transitions, which have no classical counterpart. The identification by Guo and Drake [53,54] of

a stationary solution to Schrödinger's equation for the electron-photon system opened the door to introducing the harmonic radiation field through time-independent perturbation theory and considering transitions between Volkov states with the concomitant emission of a photon. Because this formalism accounts for the entire photon field, the intensity of the fundamental optical field is incorporated explicitly into the calculations. An intriguing aspect of this framework is that the emission of only odd harmonics, the conservation of momentum, and the prediction of a shift in phase of the fundamental field during Volkov→Volkov-state transitions follow directly from the theory. The ability of an analytic expression [Eq. (51)] to describe a broad range of experimental observations supports these conclusions as well as the underlying premise that it is not necessary to attribute harmonic generation to e^--atom recombination. A consequence of quantizing the electron-photon field interaction is that, rather than having a continuous range of energies, the electron takes on discrete energies that are dictated by ATI and unchanged by harmonic generation. Furthermore, the sensitivity of theoretical predictions (harmonic polarization characteristics, in particular) to the range in j values incorporated into the calculations suggests the importance of this parameter as a diagnostic of the harmonic-generation process. The influence of a particular ATI photoelectron order on the HG process is determined by its energy and the laser pulse risetime. For laser pulse widths $\geqslant$100 fs, only the lowest two to three orders contribute substantially but decreasing the pulse width requires, as one might expect, increasingly energetic photoelectrons (i.e., higher ATI orders) to reproduce experimental spectra.

The optical field is a vital participant in the HG process as opposed to an external agent. It can be viewed as stimulating the formation of Volkov states and simultaneously contributing to the composition of the electron-photon complex or quasiatom [74]. In this sense, the photon field is loosely analogous to the background neutral gas in chemical reactions in the gas phase when, for example, transient diatomic molecules are produced by three-body collisions in which an electronically excited atom interacts with two neutral atoms.

One immediate implication of the results presented here is that measurements of ATI spectra and the polarization characteristics of individual harmonics with ultrashort optical pulses (i.e., $<$50 fs) would be valuable in further testing both semiclassical and QED theories. Furthermore, harmonic generation, in retrospect, may well prove to be a test bed for examining QED theory in a high-photon-number, nonperturbative environment.

ACKNOWLEDGMENTS

Discussions with D.-S. Guo, Y.-S. Wu, J. T. Verdeyen, and S. E. Harris as well as the technical assistance of K. Collier and the computational facilities provided by M. J. Kushner and the National Center for Supercomputing Applications at the University of Illinois are gratefully acknowledged. This work was supported by the U.S. Air Force Office of Scientific Research under Grant No. F49620-98-1-0030.

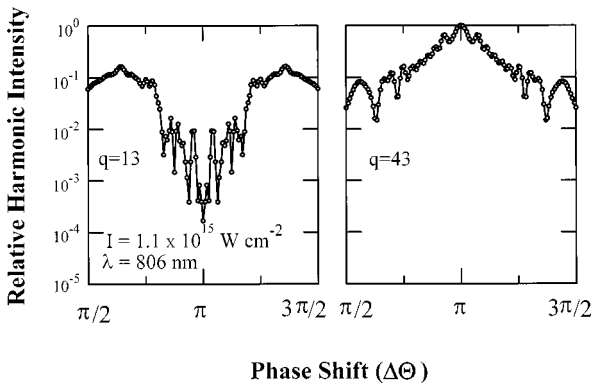


FIG. 16. Harmonic generation rate for $q=13$ and 43 in Ne as a function of the phase shift $\Delta\Theta$. The fundamental intensity and wavelength are taken to be $1.1\times10^{15}\,\mathrm{W\,cm^{-2}}$ and 806 nm, respectively. Only the lowest two photoelectron orders ($j=58,59$) are included in the calculations.

APPENDIX: DISCUSSION AND IMPLICATIONS OF THE PHASE SHIFT $\Delta\Theta$

The requirement that the fundamental optical field experiences a phase shift during Volkov→Volkov-state transitions is a key aspect of this theory. Clearly, if $\Delta\Theta=0$, then T^V [Eq. (40)] vanishes and the harmonic generation transition matrix [Eq. (22)] reflects only the production and recombination of photoelectrons. Therefore, several comments regarding the necessity for, and plausibility of, nonzero values of $\Delta\Theta$ are in order.

From a general perspective, it is well known that scattering processes require a change in the phase of the field. If, for example, the fundamental is assumed to be represented by the classical field $Ae^{i\vec{k}\cdot\vec{r}}$ prior to the interaction and $A'e^{i\vec{k}\cdot\vec{r}}$ thereafter, and the harmonic field is represented by $Be^{i\vec{k}'\cdot\vec{r}}$, then the continuity condition for the scattering process requires that

$$Ae^{i\vec{k}\cdot\vec{r}}=A'e^{i\vec{k}\cdot\vec{r}}+Be^{i\vec{k}'\cdot\vec{r}}. \tag{A1}$$

Setting $|\vec{r}|=0$ for convenience gives

$$A=A'+B. \tag{A2}$$

Energy conservation leads to the relation

$$AA^*=A'A'^*+BB^*\left(\frac{\omega'}{\omega}\right)^2=|A|^2+|B|^2q^2, \tag{A3}$$

where q is again the harmonic order. Setting $|A|=1$ and $A'=|A'|e^{i\Delta\Theta}$ and combining Eqs. (A2) and (A3) yields

$$|A'|^2+(1-2|A'|\cos\Delta\Theta+|A'|^2)q^2=1. \tag{A4}$$

Equation (A4) requires that $\Delta\Theta\neq0$ for all harmonics. Although this discussion assumes classical fields for both the fundamental and harmonics, the same principle holds for quantum fields.

The HG transition matrix [Eq. (51)] is sensitive to the value of $\Delta\Theta$ and Fig. 16 shows the calculated dependence of the harmonic generation rate on $\Delta\Theta$ for two harmonics of

806 nm ($q=13$ and 43) in Ne for an intensity of $1.1\times10^{15}\,\mathrm{W\,cm^{-2}}$. Only the lowest two ATI orders ($j=58,59$) are included in the calculations but the results are representative of those obtained under different conditions. Notice that, for both harmonics, the generation rate is symmetric with respect to $\Delta\Theta=\pi$ and that the functional behavior is dependent upon q. It is clear that the generation efficiency for a given harmonic can be optimized if a means for controlling $\Delta\Theta$ or restricting it to a narrow range can be devised.

[1] G. H. C. New and J. F. Ward, Phys. Rev. Lett. **19**, 556 (1967); see also J. F. Ward and G. H. C. New, Phys. Rev. **185**, 57 (1969).

[2] For a review of early developments, the reader is referred to J. Reintjes, C.-Y. She, and R. C. Eckardt, IEEE J. Quantum Electron. **QE-14**, 581 (1978).

[3] J. Reintjes, L. L. Tankersley, and R. Christensen, Opt. Commun. **39**, 334 (1981).

[4] J. Bokor, P. H. Bucksbaum, and R. R. Freeman, Opt. Lett. **8**, 217 (1983).

[5] A. McPherson, G. Gibson, H. Jara, U. Johann, T. S. Luk, I. McIntyre, K. Boyer, and C. K. Rhodes, J. Opt. Soc. Am. B **4**, 595 (1987).

[6] M. Ferray, A. L'Huillier, X. F. Li, L. A. Lompré, G. Mainfray, and C. Manus, J. Phys. B **21**, L31 (1988).

[7] X. F. Li, A. L'Huillier, M. Ferray, L. A. Lompré, and G. Mainfray, Phys. Rev. A **39**, 5751 (1989).

[8] N. Sarukura, K. Hata, T. Adachi, R. Nodomi, M. Watanabe, and S. Watanabe, Phys. Rev. A **43**, 1669 (1991).

[9] K. Miyazaki and H. Sakai, J. Phys. B **25**, L83 (1992).

[10] J. K. Crane, M. D. Perry, S. Herman, and R. W. Falcone, Opt. Lett. **17**, 1256 (1992).

[11] J. J. Macklin, J. D. Kmetec, and C. L. Gordon III, Phys. Rev. Lett. **70**, 766 (1993).

[12] A. L'Huillier and Ph. Balcou, Phys. Rev. Lett. **70**, 774 (1993).

[13] A. L'Huillier, M. Lewenstein, P. Salières, Ph. Balcou, M. Yu. Ivanov, J. Larsson, and C.-G. Wahlström, Phys. Rev. A **48**, R3433 (1993).

[14] Y. Nagata, K. Midorikawa, M. Obara, and K. Toyoda, Opt. Lett. **21**, 15 (1996).

[15] I. Mercer, E. Mevel, R. Zerne, A. L'Huillier, Ph. Antoine, and C.-G. Wahlström, Phys. Rev. Lett. **77**, 1731 (1996).

[16] Z. Chang, A. Rundquist, H. Wang, M. M. Murnane, and H. C. Kapteyn, Phys. Rev. Lett. **79**, 2967 (1997).

[17] Ch. Spielmann, N. H. Burnett, S. Sartania, R. Koppitsch, M. Schnürer, C. Kan, M. Lenzner, P. Wobrauschek, and F. Krausz, Science **278**, 661 (1997).

[18] Ch. Spielmann, C. Kan, N. H. Burnett, T. Brabec, M. Geissler, A. Scrinzi, M. Schnürer, and F. Krausz, IEEE J. Sel. Top. Quantum Electron. **4**, 249 (1998).

[19] M. Schnürer, Ch. Spielmann, P. Wobrauschek, C. Streli, N. H. Burnett, C. Kan, K. Ferencz, R. Koppitsch, Z. Cheng, T. Brabec, and F. Krausz, Phys. Rev. Lett. **80**, 3236 (1998).

[20] A. Rundquist, C. G. Durfee III, Z. Chang, C. Herne, S. Backus, M. M. Murnane, and H. C. Kapteyn, Science **280**, 1412 (1998).

[21] K. S. Budil, P. Salières, A. L'Huillier, T. Ditmire, and M. D. Perry, Phys. Rev. A **48**, R3437 (1993).

[22] N. H. Burnett, C. Kan, and P. B. Corkum, Phys. Rev. A **51**, R3418 (1995).

[23] F. A. Weihe, S. K. Dutta, G. Korn, D. Du, P. H. Bucksbaum, and P. L. Shkolnikov, Phys. Rev. A **51**, R3433 (1995).

[24] K. Miyazaki and H. Takada, Phys. Rev. A **52**, 3007 (1995).

[25] F. A. Weihe and P. H. Bucksbaum, J. Opt. Soc. Am. B **13**, 157 (1996).

[26] M. Yu. Ivanov, T. Brabec, and N. Burnett, Phys. Rev. A **54**, 742 (1996).

[27] M. Kakehata, H. Takada, H. Yumoto, and K. Miyazaki, Phys. Rev. A **55**, R861 (1997).

[28] Ph. Antoine, A. L'Huillier, M. Lewenstein, P. Salières, and B. Carré, Phys. Rev. A **53**, 1725 (1996).

[29] Ph. Antoine, B. Carré, A. L'Huillier, and M. Lewenstein, Phys. Rev. A **55**, 1314 (1997).

[30] M. Bellini, C. Lyngå, A. Tozzi, M. B. Gaarde, T. W. Hänsch, A. L'Huillier, and C.-G. Wahlström, Phys. Rev. Lett. **81**, 297 (1998).

[31] P. Salières, T. Ditmire, K. S. Budil, M. D. Perry, and A. L'Huillier, J. Phys. B **27**, L217 (1994), and references cited therein.

[32] J. H. Eberly, Q. Su, and J. Javanainen, J. Opt. Soc. Am. B **6**, 1289 (1989); Phys. Rev. Lett. **62**, 881 (1989).

[33] W. Becker, S. Long, and J. K. McIver, Phys. Rev. A **41**, 4112 (1990).

[34] J. L. Krause, K. J. Schafer, and K. C. Kulander, Phys. Rev. Lett. **68**, 3535 (1992); Phys. Rev. A **45**, 4998 (1992).

[35] K. C. Kulander, K. J. Schafer, and J. L. Krause, in *Atoms in Intense Radiation Fields*, edited by M. Gavrila (Academic, New York, 1992).

[36] K. C. Kulander, K. J. Schafer, and J. L. Krause, in *Super-Intense Laser-Atom Physics*, edited by B. Piraux, A. L'Huillier, and K. Rzążewski, Vol. 316 of *NATO Advanced Study Institute Series B: Physics* (Plenum, New York, 1993).

[37] P. B. Corkum, Phys. Rev. Lett. **71**, 1994 (1993).

[38] M. Lewenstein, Ph. Balcou, M. Yu. Ivanov, A. L'Huillier, and P. B. Corkum, Phys. Rev. A **49**, 2117 (1994).

[39] W. Becker, S. Long, and J. K. McIver, Phys. Rev. A **50**, 1540 (1994).

[40] S. C. Rae, K. Burnett, and J. Cooper, Phys. Rev. A **50**, 3438 (1994).

[41] M. Lewenstein, K. C. Kulander, K. J. Schafer, and P. Bucksbaum, Phys. Rev. A **51**, 1495 (1995).

[42] M. Lewenstein, P. Salières, and A. L'Huillier, Phys. Rev. A **52**, 4747 (1995).

[43] W. Becker, A. Lohr, M. Kleber, and M. Lewenstein, Phys. Rev. A **56**, 645 (1997).

[44] J. Gao, F. Shen, and J. G. Eden, Phys. Rev. Lett. **81**, 1833 (1998).

[45] C. Bula, K. T. McDonald, E. J. Prebys, C. Bamber, S. Boege, T. Kotseroglou, A. C. Melissinos, D. D. Meyerhofer, W. Ragg, D. L. Burke, R. C. Field, G. Horton-Smith, A. C. Odian, J. E.

Spencer, D. Walz, S. C. Berridge, W. M. Bugg, K. Shmakov, and A. W. Weidemann, Phys. Rev. Lett. **76**, 3116 (1996).
[46] L. V. Keldysh, Zh. Eksp. Teor. Fiz. **47**, 1945 (1964) [Sov. Phys. JETP **20**, 1307 (1965)]; F. H. M. Faisal, J. Phys. B **6**, L89 (1973); H. R. Reiss, Phys. Rev. A **22**, 1786 (1980).
[47] D. M. Volkov, Z. Phys. **94**, 250 (1935).
[48] H. R. Reiss, J. Phys. B **20**, L79 (1987).
[49] P. H. Bucksbaum, M. Bashkansky, R. R. Freeman, T. J. McIlrath, and L. F. DiMauro, Phys. Rev. Lett. **56**, 2590 (1986).
[50] U. Mohideen, M. H. Sher, H. W. K. Tom, G. D. Aumiller, O. R. Wood II, R. R. Freeman, J. Bokor, and P. H. Bucksbaum, Phys. Rev. Lett. **71**, 509 (1993).
[51] D.-S. Guo, T. Åberg, and B. Crasemann, Phys. Rev. A **40**, 4997 (1989).
[52] B. Lippmann and J. Schwinger, Phys. Rev. **79**, 469 (1950).
[53] D.-S. Guo and G. W. F. Drake, J. Phys. A **25**, 3383 (1992).
[54] D.-S. Guo and G. W. F. Drake, Phys. Rev. A **45**, 6622 (1992).
[55] J. Gao, D.-S. Guo, and Y.-S. Wu (unpublished).
[56] P. H. Bucksbaum, D. W. Schumacher, and M. Bashkansky, Phys. Rev. Lett. **61**, 1182 (1988).
[57] D.-S. Guo and T. Åberg, J. Phys. A **21**, 4577 (1988).
[58] D.-S. Guo, J. Gao, and A. H. Chu, Phys. Rev. A **54**, 1087 (1996).
[59] See, for example, C. Cohen-Tannoudji, B. Diu, and F. Laloë, *Quantum Mechanics*, Vol. II (Wiley and Sons, New York, 1977), pp. 931–934.
[60] M. Yu. Ivanov and K. Rzążewski, J. Mod. Opt. **39**, 2377 (1992).
[61] T. J. Englert and E. A. Rinehart, Phys. Rev. A **28**, 1539 (1983).
[62] See, for example, P. Kruit, J. Kimman, H. G. Muller, and M. J. van der Wiel, Phys. Rev. A **28**, 248 (1983).
[63] K. J. Schafer, B. Yang, L. F. DiMauro, and K. C. Kulander, Phys. Rev. Lett. **70**, 1599 (1993).
[64] B. Yang, K. J. Schafer, B. Walker, K. C. Kulander, P. Agostini, and L. F. DiMauro, Phys. Rev. Lett. **71**, 3770 (1993).
[65] B. Walker, B. Sheehy, L. F. DiMauro, P. Agostini, K. J. Schafer, and K. C. Kulander, Phys. Rev. Lett. **73**, 1227 (1994).
[66] B. Walker, B. Sheehy, K. C. Kulander, and L. F. DiMauro, Phys. Rev. Lett. **77**, 5031 (1996).
[67] G. G. Paulus, F. Zacher, H. Walther, A. Lohr, W. Becker, and M. Kleber, Phys. Rev. Lett. **80**, 484 (1998).
[68] C.-G. Wahlström, J. Larsson, A. Persson, T. Starczewski, S. Svanberg, P. Salières, Ph. Balcou, and A. L'Huillier, Phys. Rev. A **48**, 4709 (1993).
[69] I. P. Christov, J. Zhou, J. Peatross, A. Rundquist, M. M. Murnane, and H. C. Kapteyn, Phys. Rev. Lett. **77**, 1743 (1996).
[70] C. Kan, N. H. Burnett, C. E. Capjack, and R. Rankin, Phys. Rev. Lett. **79**, 2971 (1997).
[71] M. B. Gaarde, Ph. Antoine, A. L'Huillier, K. J. Schafer, and K. C. Kulander, Phys. Rev. A **57**, 4553 (1998).
[72] W. Becker, S. Long, and J. K. McIver, Phys. Rev. A **50**, 1540 (1994).
[73] J. Gao, F. Shen, and J. G. Eden, J. Phys. B **32**, 4153 (1999).
[74] Y.-S. Wu has recently proposed the term ''photon cloud'' to describe the electron-photon system.